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DesignFusion: Integrating Generative Models for Conceptual Design Enrichment

Conceptual design is a pivotal phase of product design and development, encompassing user requirement exploration and informed solution generation. Recent generative models with their powerful content generation capabilities have been applied to conceptual design to support designers' ideation. However, the lack of transparency in their generation process and the shallow nature of their generated solutions constrain their performance in complex conceptual design tasks. In this study, we first introduce a conceptual design generation approach that combines generative models with classic design theory. This approach decomposes the conceptual design task based on design process and design attributes, and uses the who, what, where, when, why, how (5W1H) method, function-behavior-structure model, and Kansei Engineering to guide generative models to generate conceptual design solutions through multi-step reasoning. Then we present an interactive system using a mind-map layout to visualize multi-step reasoning, called DesignFusion. This empowers designers to track the generation process and control inputs/outputs at each reasoning step. Two user studies show that our approach significantly enhances the quality of generated design solutions and enriches designer experience in human–artificial intelligence co-creation. [DOI: 10.1115/1.4065487]

Keywords: conceptual design, creativity and concept generation

1 Introduction

In today's global market, customer satisfaction has emerged as a key determinant of product success [1], user-centered product design matches design to user needs, aiming to ultimately create products that can address users' problems and meet their requirements [2]. During the entire product design process, conceptual design stands as a core stage and establishes the trajectory for subsequent detailing design [3]. It typically commences with a design problem or a set of requirements from users and culminates in one or more design solutions expressed through sketches and textual explanations [4]. The focal point of conceptual design is to address user needs and produce creative and high-quality solutions [5], which is challenging for novice designers with no training or only preliminary training. On the one hand, they lack experience in requirement analysis and struggle to accurately define user needs. On the other hand, their limited personal knowledge hinders them from fully exploring the design space to generate functional and useful solutions.

To assist novice designers in producing creative ideas, researchers have utilized data from patents, scientific publications, and other sources to construct semantic networks [6–9]. These networks calculate the semantic information between different concepts to provide designers with remotely or closely related concept stimuli. However, they often present stimuli in original form without providing explanations on how these stimuli can be applied to design targets. For example, TechNet [8] introduces related terms such as "multi-functional composite" and "rapidly convertible" for the user query "flying car," yet fails to offer a solution on achieving these functions. Lacking guidance presents a significant hurdle for users in practical application.

Recent, breakthroughs in generative models have unveiled new avenues for addressing this challenge. Large language models (LLMs) [10–12] have powerful natural language processing and

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generation capabilities and can complete various tasks such as text generation, text summarization, translation, and question answering. These capabilities stem from their extensive model parameters and the vast training data drawn from diverse sources on the Internet [11]. Consequently, LLMs offer novice designers the opportunity to access copious amounts of knowledge and have the potential to assist them in navigating the complex relationship between problem spaces and solution spaces [13,14]. And notably, LLMs can directly generate conceptual design solutions in natural language for novice designers [15,16], which do not require very detailed product specifications and performance. In addition, the capacity of image generation models [17,18] to produce remarkably realistic images using text prompts has gained public attention. They can utilize various visual concepts and seamlessly combine them to produce innovative images, showcasing significant potential in facilitating novice designers to swiftly conceive product design ideas [19,20]. Nevertheless, latest research findings have revealed three main challenges that persist during the human–artificial intelligence (AI) collaboration for addressing complex conceptual design tasks:

- **C1: LLMs cannot generate design solutions in the designerly ways of knowing.** Designers need to tackle "ill-defined" problems and propose concrete solutions [21]. The impactful solution is not simply lying there among the data but hinging on a thorough exploration of the context of the design problem and a step-by-step reasoning [3]. Since LLMs are oriented toward modeling the data distribution of language, they generate ideas by imitating the forms of existing solutions in language rather than reasoning based on meaning [22–24]. This often leads to their inability to provide substantial elaboration or further extension on these ideas, rendering the ideas lacking in practical functionality and usability [15,25].
- **C2: It's a challenge for humans to comprehend and control generated design concepts.** To explore the design space effectively, designers must know their current position in the design process, quickly identify issues, and modify the direction of design development via reformulation, analogy, or co-evolution [26]. However, most existing methods directly generate concepts or solutions [15], without offering a transparent process. Thus, inexperienced novice designers often encounter reduced controllability or even struggle to comprehend AI-generated outcomes [25,27].
- **C3: Prompt writing for generating design solutions is complex and difficult**. To fully utilize the potential of generative models for conceptual design, it is often essential to possess a profound comprehension of the design problem and the capabilities of the models, and fine-tune prompts for different tasks in the workflow of conceptual design [28]. As a result, novice designers often need to invest time and effort on prompt writing to obtain relevant and valuable responses from models [29].

In this work, we propose a novel conceptual design generation approach to guide the generative models to create solutions in the designerly ways of thinking and improve the quality of the solutions. This approach divides the conceptual design process into two key stages: requirement analysis and solution reasoning, and employs classic design methods to guide the generative models progressively through these stages. In the requirement analysis stage, the who, what, where, when, why, how (5W1H) method [30] is utilized to help an LLM conduct a comprehensive exploration of the problem space, thereby discovering the functional and emotional needs of users. In the solution reasoning stage, the function-behavior-structure (FBS) model [31] is employed to guide the LLM in a step-by-step structure design reasoning through function-behavior-structure, Kansei Engineering [32] is used to assist the LLM, and a text-to-image model in deducing appearance design solutions and images from emotional requirements. To empower novice designers with better understanding and control over the generated outcomes, we introduce DesignFusion, a conceptual design support system. DesignFusion applies the generation approach as a core workflow for designers and employs a mind-map like layout in its interface to vividly illustrate the design generation process. It allows designers to conduct fine-grained control over the inputs and outputs at each reasoning step, which enables human-led collaboration. We also present a set of prompt templates for context setting, text generation, and image generation tasks in creativity generation, which helps novice designers efficiently write prompt in creative activities and obtain output matching the objectives. These prompts are also applied in DesignFusion, simplifying the task of prompt writing for novice designers.

Through a between-subject user study, we evaluated the effectiveness of our approach. The quantitative results demonstrate that the design solutions generated by our approach have better functionality and usefulness, as well as being more aesthetically pleasing and attractive. The qualitative results prove that our approach can assist novice designers in thoroughly exploring both the problem space and solution space. A within-subject user study was also conducted to evaluate the impact of our system on designer experience in human–AI co-creation. The results show that our system significantly improves the novice designer experience in terms of transparency, controllability, collaboration, guided thinking, and confidence.

We summarize our contributions in the following:

(1) We propose a novel conceptual design generation approach.
(2) We introduce an intelligent conceptual design system, DesignFusion, to support the interaction between novice designers and the approach.
(3) We present a set of prompt templates for typical tasks in conceptual design generation.

2 Related Work

2.1 Conceptual Design. In early research [4], product design and development are typically categorized into four distinct phases: analysis of problem, conceptual design, embodiment of schemes, and detailing. Among these, conceptual design holds an essential role as it involves critical decision-making and producing novel design concepts [4]. Managing the conceptual design process presents unique challenges because "design processes seek to do something novel, once, whereas many other business processes seek to do the same thing repetitively [33]. To assist designers in gaining a more comprehensive understanding of their design process and breaking free from the limitations imposed by well-known features in existing designs, researchers have developed various process models [34].

The 5W1H method [30] assists designers in thoroughly and systematically defining and articulating design problems by asking who, what, where, when, why, and how. TRIZ [35] offers a method to support innovation by identifying and resolving contradictions in design scenarios. C–K theory [36,37] presents design as a process that moves between the knowledge space and the conceptual space. Maher and Poon [38] argued that design involves a co-evolution between problem space and solution space until mutually acceptable and compatible definitions of the problem and solution are reached. The FBS model [31] suggests that all designs involve transformations between the domains of function (describing the design's purpose), behavior (describing its function), and structure (describing what it is). Morphological analysis [39] decomposes the target function into a series of sub-functions and studies the various potential implementation methods and combinations of these methods to discover new solutions. In contrast to the aforementioned models that focus on function and structure design, Kansei Engineering [32] utilizes a mapping model between user emotions and product appearance design, translating consumers' feelings into rational design elements to achieve user-centered appearance design. Following it, many other Kansei Engineering based methods have been

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proposed to enhance the subjective quality of aesthetics [40] and improve the match accuracy between Kansei words and appearance design [2].

As mentioned above, the design and development process involve numerous interconnected issues, and these models reflect selective viewpoints on these matters. Best practices in conceptual design tasks should not be confined to any single model but rather involve integrating and applying a set of models based on the specific task at hand [34]. Therefore, to aid novice designers in product design, our approach attempts to stimulate design ideation by exploring the problem space and solution space at a macro level. Simultaneously, at a micro level for solution reasoning, the 5W1H and FBS methods, which decompose initial problems or functional requirements into several sub-problems or sequential steps for problem-solving, align closely with the chain of thought (COT) theory [41] recently highlighted in research on LLMs. COT encourages LLMs to explicitly detail the steps of reasoning and the underlying logical pathways when providing solutions, rather than merely presenting the end result. This has been shown to significantly enhance the capacity of LLMs for complex reasoning. In the realm of appearance design, existing text-to-image models can create images from concrete concept descriptions, but they struggle with abstract emotional expressions, i.e., 29% emotion accuracy [42]. Kansei Engineering, unlike other design methods, focuses on converting abstract emotional needs into concrete design elements, which can assist text-to-image models in overcoming this challenge. Therefore, our approach leverages the principles of the 5W1H method, the FBS model, and Kansei Engineering to guide step-by-step reasoning for problem analysis, structure design, and appearance design.

2.2 Creativity Support for Conceptual Design. Knowledge serves as a pivotal resource for conceptual design ideation in product design and development [43,44]. Successfully completing the design process necessitates building upon a substantial foundation of existing knowledge. Yet, designers often possess limited knowledge. Scientific publications and patents usually contain a wealth of cutting-edge scientific and technological knowledge. Many studies [7,8,45,46] attempt to extract knowledge from scientific publications and patents to support design ideation. Some studies turned to utilizing the information in encyclopedias for design tasks [9,47], which covers a broader spectrum of knowledge. All these approaches share a common characteristic of building semantic networks based on patent, research paper, or encyclopedia data. By computing the semantic distances between design targets and knowledge, they offer design stimuli or knowledge insights to both designers and even non-experts users. In addition, some studies [48,49] use more intuitive imagery to provide designers with visual stimuli within group brainstorming.

With the rapid advancements in deep generative models, researchers have moved beyond the confines of retrieval-based methods for providing creative stimuli. They are now exploring the potential of utilizing these generative models to directly generate easily comprehensible and creative design concept [50,51], thereby taking creativity support to the next level. LLMs like ChatGPT [10], Llama [12], and bert [52] have extensively learned from massive amounts of textual data and possess robust natural language processing capabilities, including tasks such as text summarization, text classification, and text generation. To assist conceptual design, researchers introduces LLMs to identify requirements [53], extract design knowledge [14,54], and generate design solutions [16]. Zhu et al. [15] presented several data-driven approaches to control reasoning (problem driven or analogy-driven) when finetuning GPT-2 and GPT-3 for design concept generation. However, recent research [25] indicates that due to the lack of transparency and control in the model's generation process, this kind of approach faced challenges in yielding effective outcomes. To enhance the interpretability and quality of content generated by LLMs, methods such as AI chain [23] and chain-of-thought [55] emerged. These methods involve task decomposition and step-by-step reasoning to address complex tasks. PopBlends [56] achieved concept blending tasks through two rounds of divergent and convergent process with GPT-3. Wang et al. [57] extended this notion to design tasks, guiding GPT-3 to generate design solutions by following the logical steps of FBS. However, this method focuses solely on product structure design, overlooking the significance of appearance design as a crucial aspect of conceptual design.

Some researchers also apply image generation models to synthesize product images and provide designers with more direct inspirations. Chen et al. [58] applied semantic networks for seeking creative concepts and used GANs to generate product images of the combination of two concepts. Quan et al. [59] combined Kansei Engineering and neural style transfer systems to achieve customized modifications to existing product appearances. The latest image generative models such as stable diffusion [17], Midjourney [60], and DALL-E 2 [18] leverage CLIP [61] to achieve alignment between text and images, enabling them to generate captivating images based on user-input text prompt. 3DALL-E [19] integrates GPT-3, DALL-E, and CLIP into a cad software to help designers craft text prompts with design language and generate images for inspiring the rendering of product cad models.

The above creativity support methods have preliminarily verified the feasibility of using generative models for design, but their end-to-end generation relies on the learning of the existing solution form instead of meaning, lacking adherence to design thinking principles and control over the results. They also focus on a local part of conceptual design and neglect a comprehensive discussion of the entire process. Our work addresses this gap and provides a holistic approach in this regard.

3 Method: Conceptual Design Generation With Generative Models

This section illustrates our proposed conceptual design generation approach and the prompt templates. The approach transforms the designerly way of thinking into the conceptual design generation by analysing complex design problems with the 5W1H method and generating high-quality design solutions according to the FBS model and Kansei Engineering.

3.1 Approach for Generative Conceptual Design. We hypothesize that breaking down a design challenge into discrete sub-tasks, with guided directions for reasoning in each sub-task, could enhance the model performance and extend this performance to the overarching design task [23,24]. Building on this premise, we introduce a novel approach, as shown in Fig. 1, to decompose conceptual design tasks in terms of horizontal design process and vertical design attributes.

Our approach originates from a design problem and proceeds through two sequential stages: requirement analysis and solution reasoning [3,4], ultimately leading to feasible design solutions that meet the specified criteria. The design problem constitutes textual input supplied by novice designers, outlining the requirements from users, which is the issue to be addressed. Typically, designers need to analyze problems and make a thorough and organized statement to better understand the problem. When introducing an LLM to help designers analyze problems, extracting detailed and comprehensive requirements directly from the initial design problem is a challenge for it. To address this, the 5W1H method [30] is introduced to break down and gain insights into the design problem. An LLM is tasked with answering six questions based on the design problem: *Who will use the product, When and where will it be used, What will it be used for, How will it be used, and Why will it be used?* After an in-depth exploration of the problem and its context, more comprehensive user requirements can be distilled from the answers. The user requirements can typically be categorized into two primary types: functional requirements and emotional requirements [2,62]. Functional requirements define the user's expectations for the purpose of the product,

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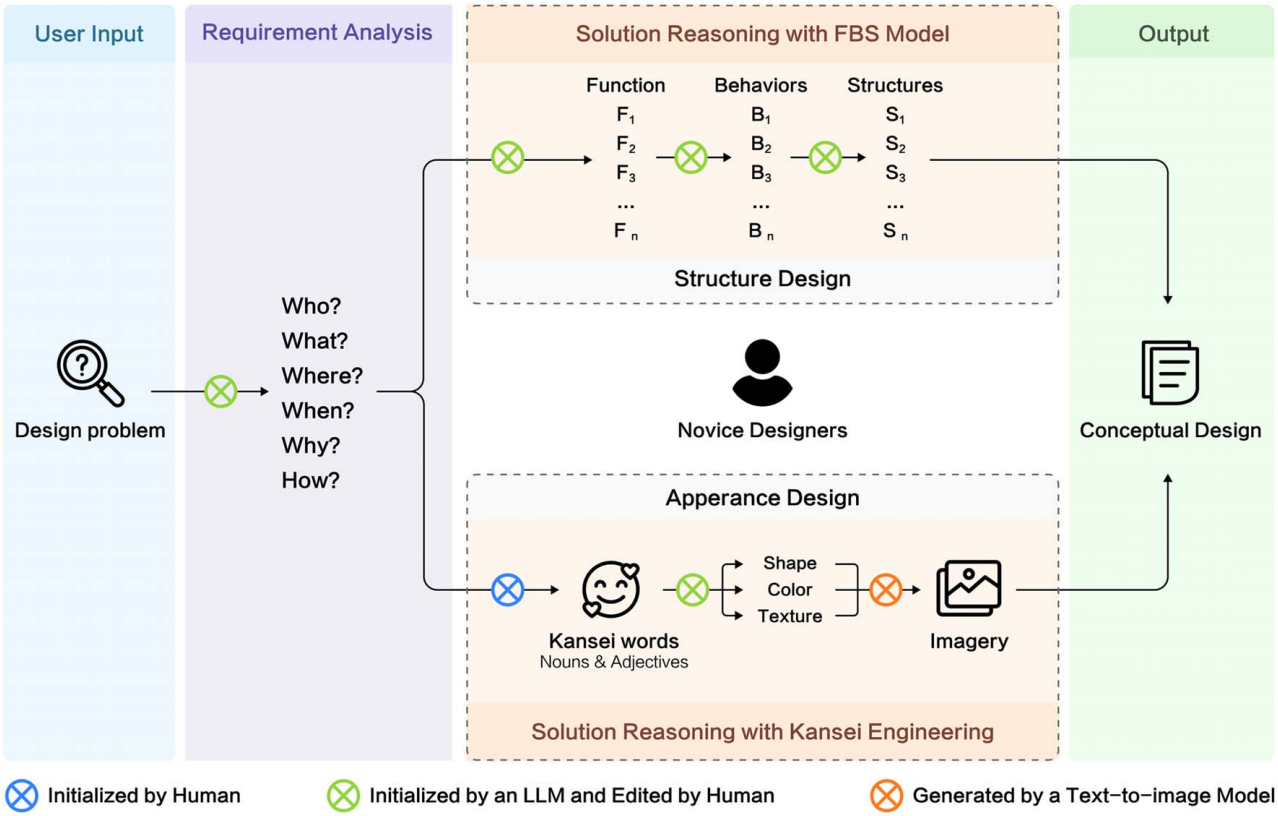


Fig. 1 An overview of the conceptual design generation approach

which can be achieved through structure design [31,63]. Emotional requirements represent the emotions or feelings that users expect the product to convey, usually addressed through appearance design [32]. Therefore, to make the LLM more effective and targeted in addressing different design tasks with varying attributes, we categorize conceptual design into two parts: structure design and appearance design, and propose the following two solution reasoning processes based on their respective characteristics.

3.1.1 Structure Design Process. Product structure represents the elements of products and their relationships [63]. The structure design needs to meet functional requirements, but the direct transformation from functional requirements to structure design is difficult. The FBS model [31] represents this process as a set of transformations between three fundamental constructs—function, behavior, and structure. Taking inspiration from it, we implement the reasoning for structure design in two steps: transformations from function to behavior, and from behavior to structure, where behavior is interpreted as the product performance expected to achieve the desired function. These two steps are executed by the LLM, i.e., target functions are input into the LLM to ask which behaviors could enable these functions (as shown in Fig. 2(*b*)), and the transformation from behaviors to structures is similar. After each transformation, designers filter the generated content within the problem context. Especially when the structures are generated, they check whether the "actual" product performance aligns with the "expected" behavior, adjusting to structures or iterating through these steps as necessary until reasonable outcomes are achieved. Ultimately, designers select and integrate the generated structural components to obtain the final proposal for structure design.

Appearance design process. Product appearance design requires the designer to create and combine visual design elements. Creating visual elements that respond to user emotional requirements is a core but challenging part of this. To capture user emotional requirements, our approach leverage Kansei words to represent them based on Kansei Engineering [32]. We require the designers to identify one or more Kansei words in line with the design problem, which range from adjectives articulating user expected feelings, such as "vivid" or "warm", to nouns representing inspiration sources or imaginary like "cloud" or "impressionism". Then we innovatively utilize the LLM to model the relationship between Kansei words and visual design elements. To enhance the LLM's performance, product visual elements are categorized into shape, color, and texture following aesthetic design principles [64]. The LLM takes the Kansei words as the stimulus and outputs precise values and explanations for these three types of design elements. For instance, the LLM produces "warm yellow" as the color and "plush" as the texture when given the input "warm". The designers then make their selections from the output elements. Subsequently, a text-to-image model is integrated to combine and transform these individual and textual design elements into more intuitive graphic representations. This allows designers to gain a deeper understanding of the generated appearance design solutions, and speed up the optimization and iteration of it.

In the aforementioned process, the output of each sub-task is the result of human–AI co-creation. The designer can not only review and revise the content generated by the generative models for each sub-task but also creatively expand the design space, drawing inspiration from this content. The modified output will serve as the input for the next step reasoning of the generative models. This extends the designer's involvement into every step of conceptual design creation, amplifying the exchange of ideas and interaction between the designer and the generative models, and facilitating the designer's control over the design direction. Moreover, when problems arise in the final output solution, designers can trace back along the design process to quickly find the root cause of the problem and rapidly iterate it.

3.2 Prompt Engineering. Prompts frequently have a significant influence on the output of generative models. With the right set of prompts, an LLM trained extensively on diverse raw text data in an entirely unsupervised manner can address a wide range

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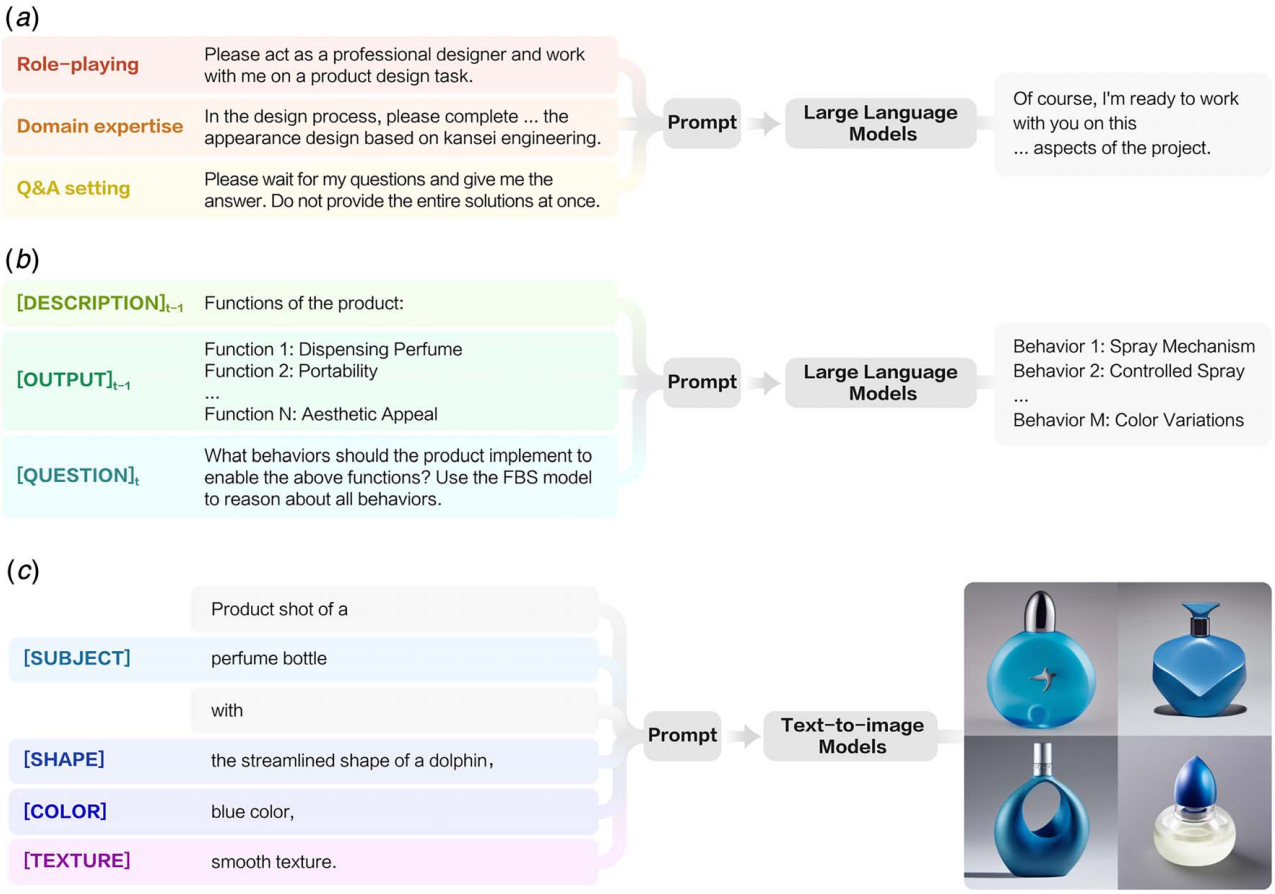


Fig. 2 Examples for each type of prompt in our approach: (*a*) the prompt for context setting, (*b*) the prompt for text generation task, and (c) the prompt for image generation task

of tasks with zero-shot prompts. However, this method brings forth the necessity of prompt engineering, which involves selecting the most fitting prompts for the specific task at hand [24]. To improve the efficiency of designers using our approach and elevate the quality of generated design solutions, we have undergone multiple iterations and testing phases to formulate a set of prompt templates designed for our approach. These prompt templates can be categorized into three types: (a) the prompt for context setting, (b) the prompt for text generation task, and (c) the prompt for image generation task. Prompts (a) and (b) are applied to LLMs, and prompt (c) is applied to text-to-image models. Examples for each type of prompt are provided in Fig. 2. The complete details of the prompts for our approach can be found in Appendix A.

3.2.1 The Prompt for Context Setting. The prompts utilized for context setting include role-playing, domain expertise, and question-and-answer setting. Role-playing prompting methodology has been previously proven to significantly enhance LLMs' performance under zero-shot settings across summarization evaluation [65], arithmetic [66], and social activities [67]. Diverging from prior methods that directly concatenate role-playing and questions within a single prompt, our approach inputs the role-playing prompt to the LLM before the long-term dialog interaction between the designer and the LLM. This prompt instructs the LLM to embody the persona of a professional designer. Early-stage immersion aids the LLM in achieving enhanced performance throughout the entirety of the generation. Domain expertise in the prompts informs the LLM to generate results based on the given classic design methods. Furthermore, to facilitate stepwise reasoning and prevent the generation of unnecessary replies, the context-setting prompt also pre-specify the question-and-answer mode for the entire dialog. As shown in Fig. 2(*a*), when the context setting prompt is inputted to an LLM, it provides a confirmation response.

3.2.2 The Prompt for Text Generation Task. After establishing the context, prompt chaining [23] is employed to complete the process of human–AI co-creation. This method involves progressively posing questions to the LLM for each sub-task, with the output of the previous step embedded within the prompt inputted into the LLM for the subsequent step. It assists the LLM to answer the current question based on the output of the previous sub-task, and can be applied to task-decomposed workflows to ensure logical coherence between consecutive steps [68]. Consequently, the prompt template that directs the LLM for the tth step sub-task within our approach can be defined as follows:

$$[DESCRIPTION]_{t-1}: \\ [OUTPUT]_{t-1}, \\ [QUESTION]_t \tag{1}$$

Here $[DESCRPTION]_{t-1}$ provides an explanatory account of the previous task, $[OUTPUT]_{t-1}$ denotes the results of human–AI collaboration in the previous step, and $[QUESTION]_t$ outlines the question of the current sub-task that the LLM needs to address. An illustrative example of the prompt for executing the function-behavior reasoning is showcased in Fig. 2(*b*). This prompt first outlines the functions that the product should have, derived from the prior step, and then asks the LLM what behaviors the product needs to take in order to achieve these functions.

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3.2.3 The Prompt for Image Generation Task. Recent empirical work [28] has demonstrated that prompts used for image generation often require descriptions including both the main subject of the image and its style, as "SUBJECT in the style of STYLE." However, it lacks explicit definitions about appearance attributes when applied to generating visual representations for product design. Aligning with the three primary attributes of product appearance outlined in our approach, we refined the prompt template for appearance design as follows:

$$\textit{Product shot of a } [\textit{SUBJECT}] \textit{ with } [\textit{SHAPE}],\ [\textit{COLOR}],\ [\textit{TEXTURE}] \tag{2}$$

Here *Product shot* defines the intended image style, [*SUBJECT*] offers a product description which can be populated with the product's name, while [*SHAPE*], [*COLOR*], and [*TEXTURE*] can be filled with visual design elements representing the product's shape, color, and texture, respectively. Figure 2(*c*) illustrates an example of such a prompt and the corresponding image generated by stable diffusion, aimed at designing a perfume bottle with the streamlined shape of a dolphin, blue color, and smooth texture.

4 User Study I: Evaluating the Performance of Conceptual Design Generation

In order to evaluate the effectiveness of our conceptual design generation approach, we conducted a user study involving 30 participants to compare the performance of our approach and two baseline approaches on a conceptual design task. An overview of this user study is provided in Fig. 3.

4.1 Evaluation Method

4.1.1 Baselines. Since conceptual design is the process of producing new solutions, we conducted a between-subject rather than within-subject study to avoid the potential influence of cognitive fixation that could occur when designers repeatedly complete the same design task. All participants were divided into three groups labeled as the *Group A (fixed prompt)*, *Group B (free conversation)*, and *Group C (our approach)*. They were all given the same conceptual design task to complete. Participants in Group A employed a method similar to existing work, where they used an LLM to directly generate design solutions with a standard prompt for the given design problem and used these generated solutions as inspiration for their ideation. The standard prompt was provided by us and can be seen in Fig. 3 and Appendix A. Participants in Group B engaged in free-form conversations with an LLM and produced design solutions based on the content of these conversations. Participants in Group C utilized our proposed approach to gradually complete the requirement analysis and the reasoning for the structure and appearance design with an LLM, using prompts provided by us based on the prompt templates (Sec. 3.2) at each step, more details can be seen in Appendix A. The underlying LLM and text-to-image model for all three group are the same: GPT-3.5 for the LLM and stable diffusion 2.1 for the text-to-image model. Interactions between participants and GPT-3.5 occurred using the official interactive tool ChatGPT provided by OpenAI, while interactions with stable diffusion 2.1 took place using the online tool [69] provided by stability-AI.

4.1.2 Participants. We openly recruited novice designers as participants on a university social networking site, with recruitment criteria requiring participants to have over 3 years of education experience in design or mechanical engineering, experience with ChatGPT and Stable Diffusion, and the ability to communicate fluently in English. We recruited a total of 33 candidates, and after review, 30 met our standards (age range 20–31, $M = 22.63$, $SD = 2.54$). We randomly divided the participants into three groups. Each participant was paid $10/h, and the average time to complete the design task was about 40 min.

4.1.3 Task and Procedure. In this experiment, each participant was asked to complete a product conceptual design task based on the content generated by GPT-3.5 and Stable Diffusion. The specific design task given to participants involved designing *a lounge chair intended for domestic use that needs to be appealing and attract the user to sit and relax.* Chair design is a classic and multifaceted topic in product design, involving considerations related to functionality, aesthetics, ergonomics, material selection, and more. Additionally, domestic lounge chairs often require customization to meet the unique preferences of individual users. This task served as a typical representation of the product design challenges our work aimed to address.

Before the experiment started, each participant received a 30-minute training on prompt writing for text and image generation and a 10-minute training on how to effectively utilize ChatGPT and Stable Diffusion. Specifically, during the image generation prompt training, we provided participants with a detailed introduction to the tips for adjusting the weighting of different keywords within the prompt and the fundamental elements within a prompt [28,70], including image style, the main subject of the content, scene details, and image quality. Following the training, we introduced the experimental method and the conceptual design task to each participant and the formal experiment began. During the experiment, participants were not allowed to use any tools other than those designated. Specifically, Group A was prohibited from inputting any content other than the provided standard prompts to GPT-3.5. In cases where participants were dissatisfied with the generated output, they were allowed to repeat the generation process multiple times. Ultimately, each participant was tasked to create both a textual description of the product's structure and an image illustrating the product's appearance design features, culminating in the submission of one completed design solution. Notably, the image did not need to explicitly represent the product's structural features. The entire experiment was conducted under the observation of a researcher to prevent participants from engaging in actions that deviated from the experimental rules. Upon completing the experiment, each participant was invited to provide written responses about the challenges they encountered while collaborating with AI in the design process, as well as whether these challenges were resolved.

4.1.4 Data Collection and Evaluation. We collected and analyzed two sets of data. The first set of data consists of expert ratings for the conceptual design solutions created by the participants. To conduct a detailed and fair evaluation of these proposals, we referred to the conceptual scoring method proposed by Ulrich et al. [3], specified multiple evaluation criteria and selected a reference conceptual design solution as a benchmark. Three experienced design experts were invited to rate 30 solutions relative to the reference design, with each solution created by an individual participant. Two of these experts are professional designers in the home product design industry with more than 6 years of design experience. The third expert is an industrial design researcher with 5 years of working experience (none of the experts are the authors). For structure design, we considered three criteria including novelty, functionality, and usefulness following Refs. [71,72]. Regarding aesthetic design, we considered six criteria including balance, proportion, simplicity, unity, harmony, and attractiveness. The first five criteria were selected based on commonly used aesthetic design principles [64], while attractiveness assessed whether the appearance design met the emotional requirement stated in the task description, "be appealing and attract the user to sit and relax." A detailed description of each criterion is provided in Appendix B. We chose a design named "ELMER'" [73] from Red Dot Winner 2023 as the reference design. The detailed information about ELMER is provided in Appendix D. The experts were requested to assess the performance of each solution in each criterion compared to the reference design using a five-point Likert scale [3]

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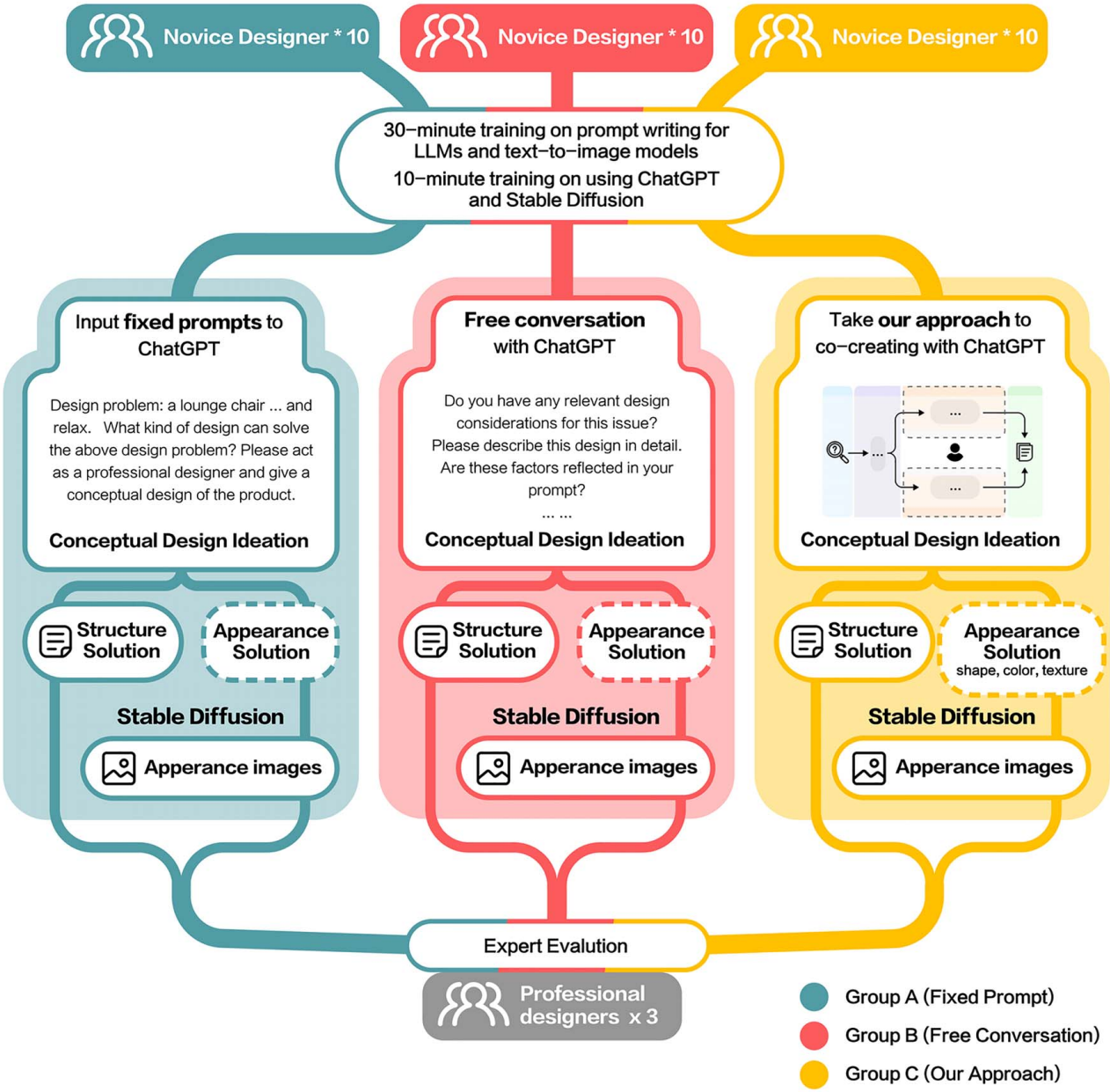


Fig. 3 An overview of the procedure for user study I

(1: much worse than reference, 2: worse than reference, 3: same as reference, 4: better than reference, 5: much better than reference).

The second set of data comes from the interactions between the participants and GPT-3.5. We collected all the text generated by the LLM in the experiment and manually extracted keywords representing structure solutions and appearance solutions. We also extracted keywords from the final concept designs submitted by the participants using the same method. To evaluate the quality of design solutions generated by the LLM and whether they were conducive to the designers' ideation, regarding the keywords in designer's solution as positive samples, we calculated the precision rate and recall rate of the LLM-generated keywords with the following equations, where G_k represents the LLM-generated keywords, D_k represents the designer's keywords, and O_k represents the keywords overlapping between the two

$$Precision = \frac{O_k}{G_k} \tag{3}$$

$$Recall = \frac{O_k}{D_k} \tag{4}$$

The keyword extraction process was carried out independently by two authors. The generated results from different groups were randomly shuffled before extraction to eliminate bias. After individual completion, any disputed portions were collectively discussed, and the results were determined. Multiple keywords with similar meanings were manually consolidated into a single keyword during the extraction process.

4.2 Quantitative Results. Tables 1 and 2 respectively present the evaluation results from experts for the structure and appearance design solutions produced by three groups of designers. Figure 4 illustrates the score distribution for three groups of designers. To improve the visualization of identical scores within a group, we adjusted the transparency of each scatter plot point to an opacity level of 0.3. The intensity of the color signifies the degree of overlap, with darker shades indicating a greater accumulation of

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Table 1 The results showing average scores and standard deviations of the structure design solutions of each group in user study I

Group	Novelty	Functionality	Usefulness
Group A (FP)	**3.70** ± 0.71	3.97 ± 0.55	3.47 ± 0.71
Group B (FC)	3.20 ± 0.55	3.60 ± 0.56	3.57 ± 0.42
Group C (Ours)	3.33 ± 0.42	**4.13** ± 0.45	**3.77** ± 0.39

Note: FP represents fixed prompt and FC represents free conversation. Bold indicates the best performance, and underline indicates the second best.

points. In terms of structure design, solutions from Group C marginally surpass those of the other groups in both functionality and usability, as reflected by their slightly higher average scores. Additionally, the scatter plot illustrates a higher concentration of scores within Group C, indicating more consistent performance. But the Group C scored lower in novelty compared to the Group A, which was also noted by four designers from Group C after the experiment, mentioning that the content generated by the LLM lacked sufficient innovation. The varying performance among the three groups can be attributed to the different methods employed in concept design generation. The Group A relies on the LLM to directly generate solutions, lacking the process of requirement analysis. Consequently, their designed products often fail to meet user needs in terms of functionality. In addition, according to LLMs' training and generation principles [11], they generate solutions by modeling the distribution of various language tokens, predicting the probability of each token, and randomly sampling from it. Through multiple samplings, LLMs can generate a variety of solutions, including some innovative ideas, but usefulness heavily depends on the model's performance on prediction. For example, after reviewing several candidate solutions generated through repeated inputs of the fixed prompt, a designer from Group A selected a satisfactory option and conceptualized a "puzzle lounge chair." This concept was widely regarded by experts as highly innovative, each awarding it a perfect score of 5 for novelty (the highest data point for Group A). However, this concept received the lowest score in usability, a mere 1.66 (the lowest data point for Group A), highlighting a disconnect between the creative concept and its practical feasibility. In complex design tasks, as the difficulty of the task increases, the accuracy of predictions decreases [23,24], thereby diminishing the usefulness of the generated solutions. Conversely, Group C leverages the generation approach we proposed. This approach systematically explores the problem space using the 5W1H method, assisting designers in comprehensively and accurately capturing product functional requirements. Consequently, they propose solutions with improved functionality. Additionally, the approach aids the LLM and novice designers in developing designerly ways of knowing, progressing from function to behavior to structure incrementally. This reduces the difficulty of model inference and prediction at each step, thus enhancing the accuracy of the inferred results and ensuring the practical usefulness of the final structure solutions. This increased inference accuracy naturally impacts the novelty of the results, but our approach strikes a balance between the two. Group B performed relatively poorly across all criteria. This is primarily due to the lack of design and prompt writing experience among novice designers. Based on feedback from designers after the experiment, seven designers from Group B indicated that they encountered challenges with formulating prompts, including uncertainty on how to articulate the tasks they needed the LLM to accomplish and instances where the LLM failed to generate responses aligned with the requirements they had input, while designers from Groups A and C rarely mentioned encountering such issues. Many of them did not resolve this issues and ultimately chose to make the LLM directly provide design solutions without systematically exploring, but fails to fully leverage the generative capabilities of the LLM, resulting in poor performance on each criterion.

In terms of appearance design, Group C achieved the highest average scores across all criteria. In addition to the previously mentioned improvements in model performance, this success can also be attributed to our approach's focus on user emotional requirements and the structured representation of appearance design. The appearance solutions and visual images derived from the Kansei words tend to show enhanced attractiveness, while the other two groups might have placed less emphasis on meeting the emotional requirements of users, leading to their comparatively poorer performance. Furthermore, in our approach, appearance design of products is structured into three kind of visual elements: shape, color, and texture. All visual elements' selection is based on the Kansei words, ensuring the unity of design elements and harmonious expression of colors. In contrast, the other groups' individual solutions relied on different rule for selecting various visual elements, resulting in inconsistency in style, disrupting overall harmony and unity. The introduction of numerous visual elements also led to a lack of simplicity. Moreover, our proposed structured representation and the product image generation template enhanced the quality of the prompts inputted into the text-to-image model. This helps the designers obtain product images that better matched the appearance design solutions.

Figure 5 illustrates the precision and recall of LLM-generated keywords across the three groups, regarding the keywords from the designers as positive samples. Group C achieved the highest precision and recall for both structure and appearance design ($R_{str} = 95.62\%$, $P_{str} = 80.13\%$, $R_{apr} = 85.21\%$, $P_{apr} = 75.09\%$). This implies that most of the content generated by our method with the LLM was effectively incorporated by the designers, and the majority of the ideas in the designers' solutions originated from the generated-content. Combining the results from Tables 1 and 2, we can infer that our method has the capability to generate solutions that are more satisfying to designers and of higher quality. On the contrary, Group A exhibited the lowest precision and recall ($R_{str} = 48.58\%$, $P_{str} = 34.78\%$, $R_{apr} = 69.44\%$, $P_{apr} = 33.33\%$). This can be attributed to that the solutions directly generated by the LLM contained a wide range of creative ideas but lacked detailed explanations on implementation or application, leading to many ideas not being adopted by the designers. Group B, having the ability to freely interact with LLM, allows designers to inquire about unclear or problematic issues, guiding the LLM to generate more detailed and feasible solutions, resulting in relatively higher precision and recall ($R_{str} = 75.00\%$, $P_{str} = 63.15\%$, $R_{apr} = 81.39\%$, $P_{apr} = 59.88\%$).

4.3 Qualitative Results. Figure 6 shows the concept solution examples for each group. It can be seen that our approach assists designer and the LLM to propose higher-quality solutions that better fulfill user needs during subsequent reasoning stages. For

Table 2 The results showing average scores and standard deviations of the appearance design solutions of each group in user study I

Group	Balance	Proportion	Simplicity	Unity	Harmony	Attractiveness
Group A (FP)	2.73 ± 0.93	2.70 ± 0.82	2.67 ± 0.67	2.63 ± 0.78	2.77 ± 0.75	2.70 ± 0.88
Group B (FC)	2.97 ± 0.56	3.03 ± 0.60	2.73 ± 0.49	2.67 ± 0.47	2.97 ± 0.71	2.90 ± 0.65
Group C (Ours)	**3.07** ± 0.72	**3.10** ± 0.72	**3.20** ± 0.86	**3.27** ± 0.90	**3.33** ± 0.68	**3.40** ± 0.94

Note: FP represents fixed prompt and FC represents free conversation. Bold indicates the best performance, and underline indicates the second best.

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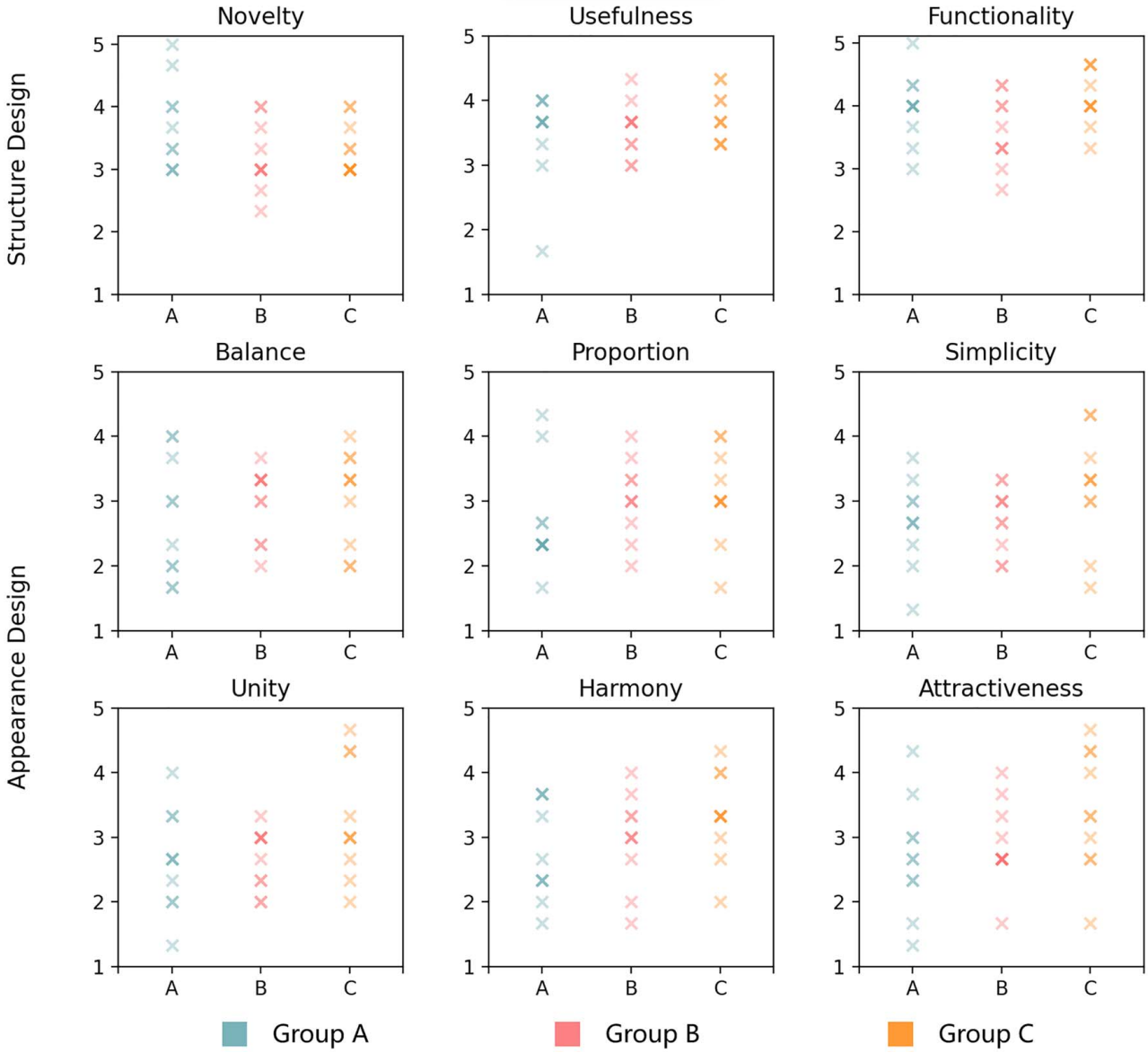


Fig. 4 Distribution of scores for the structure and appearance design solutions of each group in user study I

instance, the structure solution of Group C considers safety and users' activities like reading, suggesting "*a wide and stable base, rubber or non-slip pads, and reinforcement at stress points*" and "*integrated accessories: side tables, cup holder, or pockets.*" In contrast, the structure solution of Group B adequately addresses the foundational design but lacks novelty and functionality. Group A presents a highly creative idea for "*a modular puzzle chair,*" yet fails to provide practical implementation plans or further details.

Regarding appearance design, it is evident that Group C's starting point with Kansei words such as "cloud-like," "warm," and "relaxing" led to a more attractive and holistic solution describing shape, color, and texture. This solution better reflects the emotional requirements of the design task, which demand an "appealing" chair that invites users to sit and relax. In contrast, the solutions of the other two groups are less specific and fail to consider emotional requirements adequately. Consequently, the generated images exhibit weaker aesthetic performance and attractiveness.

5 System Design and Implement

The results of user study I have demonstrated the effectiveness of our proposed approach in generating conceptual designs. To further enhance the interaction between designers and our approach and improve the user experience in human–AI co-creation, we develop a system, DesignFusion, using this approach as the workflow. In this section, we introduce the design objectives, interface design, and implementation details of DesignFusion.

5.1 Design Objectives. As shown in Fig. 1, designers are at the center of the conceptual design generation approach, responsible for inspecting and controlling the results of each step of reasoning in the design process. This becomes particularly crucial when the final design solution fails to meet requirements, allowing designers to backtrack the design process, identify issues, and iterate on solutions. However, most existing systems such as ChatGPT and Midjourney primarily support user interactions with generative models in a linear conversational manner, limiting flexibility in user interactions and quick solution iteration. According to our approach's features and the drawbacks in existing interactive systems observed in user study I, we have outlined three key design objectives for DesignFusion's interface as follows:

O1: Provide an intuitive and visual representation of the conceptual design generation approach. Our approach divides the conceptual design into two parts, with each part

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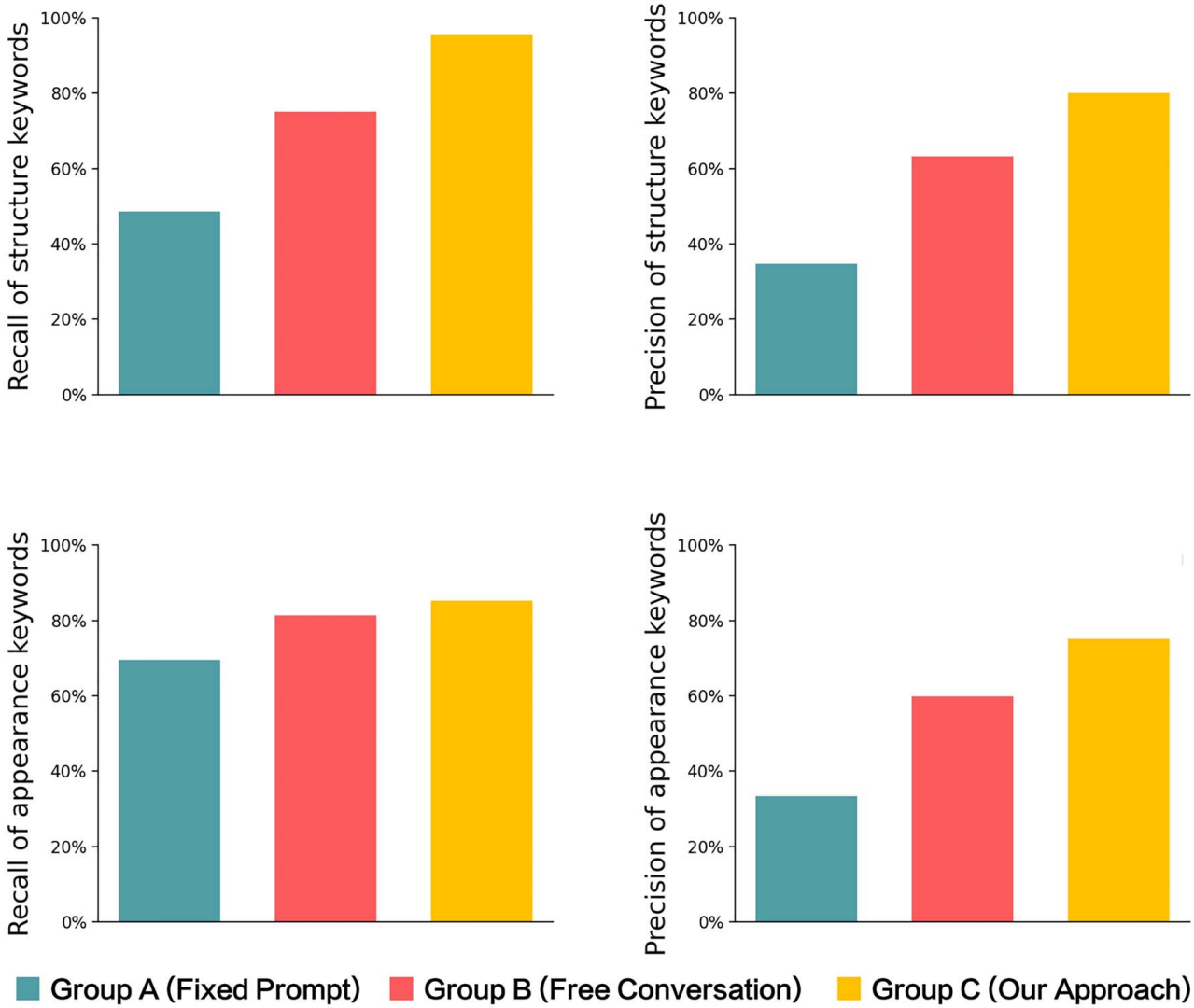


Fig. 5 The precision and recall of LLM-generated structure and appearance keywords in the three groups for user study I

comprising multiple sub-tasks for requirements analysis and solution reasoning. There are complex relationships among these sub-tasks, such as parallelism and asynchronism. This calls for an intuitive interface that visually reflects the relationship between them, aiding users in understanding and tracking the design process.

O2: Ensure that designers can operate precise control over the output of each sub-task. To enhance the transparency and controllability of the approach, the interface must ensure that designers can access the output of each reasoning step and have the capability to flexibly modify it. Moreover, when designers engage in an iterative design, the interface should support fine-grained local modifications to avoid altering the entire solution and thereby improve iteration efficiency.

O3: Integrate textual idea generation and visual idea generation into a unified system. Visualizing potential ideas as images can assist designers in rapidly evaluating and iterating. Existing interaction systems mostly support either textual idea generation or visual idea generation separately, leading to negative impacts on the designer's experience due to frequent system switching. Therefore, we aim to consolidate these functionalities into a single system.

5.2 Interface and Interaction Design. Based on the above design objectives, we design and develop a user interface as depicted in Figs. 7 and 8. To clearly showcase the entire conceptual design generation approach, we divide the user interface into two pages, one dedicated to structure design and the other to appearance design. These two pages can be navigated with the *Tab bar* (A). Each page comprises a *Requirement View* (Fig. 7(*b*)/Fig. 8(*e*)), a *Reasoning View* (Fig. 7(*c*)/Fig. 8(*f*)), and a *Solution View* (Fig. 7(*d*)/Fig. 8(*g*)). Designers initially perform requirement exploration and structure design on the first page, then transition to another page for appearance design. On each page, they follow a sequential flow from left to right, progressively conducting requirements analysis and solution reasoning, culminating in the selection of the final solution **(O1)**.

5.2.1 The Structure Design Page. As shown in Fig. 7, designers start by inputting the product design problem into the *Requirement View* (b1). After clicking the *Generate Requirements* button, the LLM generates the answers to 5W1H: *Who*, *What*, *Where*, *When*, *Why*, and *How* (b2). The answers are displayed in the textbox below each question. Designers can review and edit the answers based on their insights **(O2)**. Once they confirm with the answers, they can click the *Generate Functions* button (b3) to request the LLM to generate the functional requirements based on the answers and proceed to the solution reasoning stage. In the *Reasoning View* (C), to visually represent the solution reasoning process in structure design, a mind map is employed. The root node of this mind map symbolizes the target product name, with

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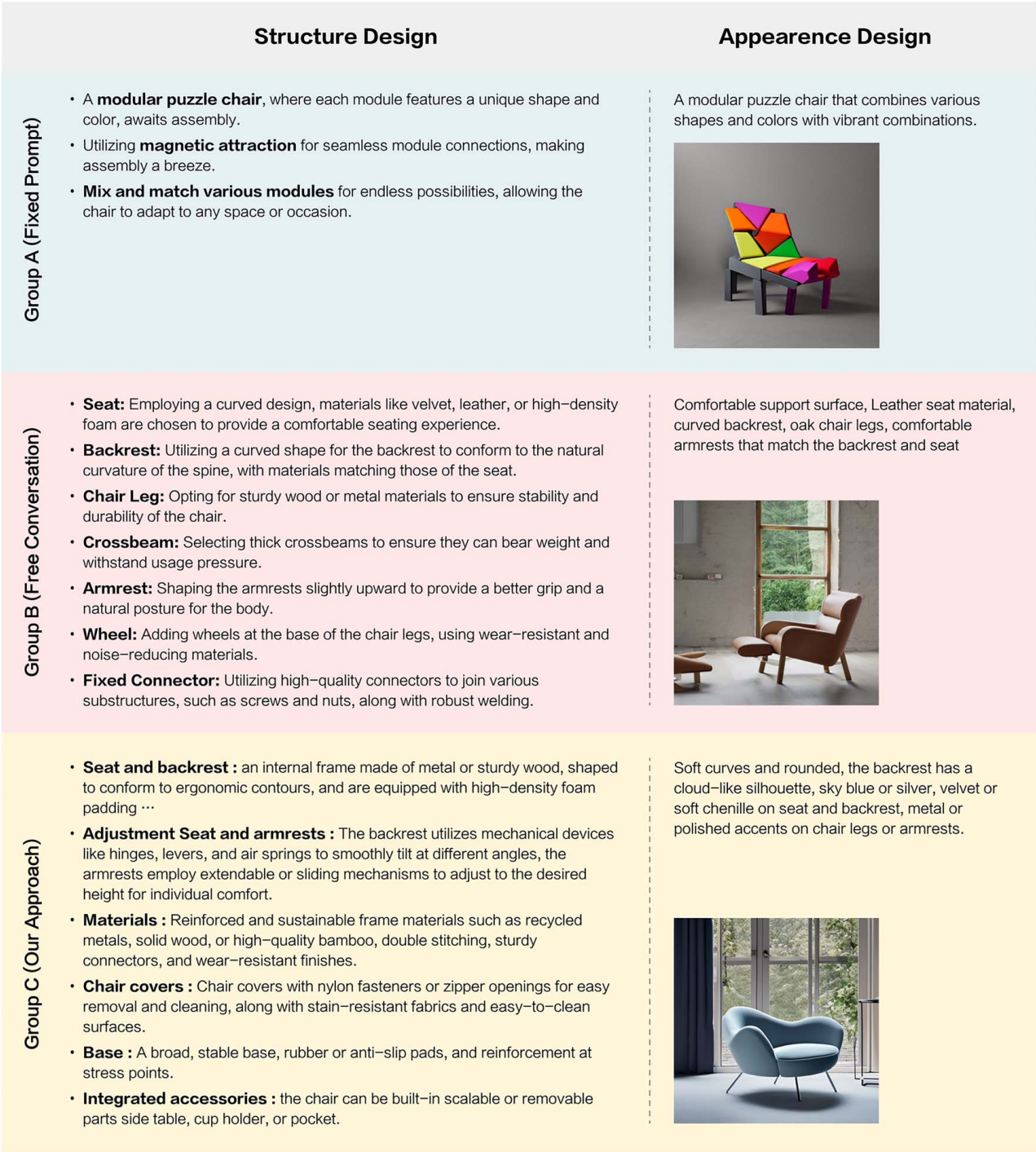


Fig. 6 Examples of concept design solutions produced by the three groups in user study I

the first-level child nodes below it (c1) representing the expected product functions derived from the answers. Subsequently, designers can click the *Generate Behaviors* and *Generate Structures* buttons (b3) to sequentially obtain the product behaviors (the second-level child nodes, c2) and structures (the third-level child node, c3) as reasoned by the LLM. Each function can deduce one or more behaviors, and each behavior can deduce one or more substructures capable of implementing that behavior. The parent–child relationships in the mind map offer a vivid depiction of the forward derivation from functions to behaviors and from behaviors to structures. They also aid designers in backward tracing their reasoning steps during the iteration, facilitating the rapid identification of issues **(O1)**. For any node within the mind map, designers can undertake four operations: edit the node, delete the node, append a child node, and regenerate child nodes. This versatility empowers designers to precisely govern the outcomes at each step during forward reasoning. During backward tracing, it allows for localized modification, sparing the need for an entire solution regeneration and thereby expediting the iterative process **(O2)**. Upon completing the solution reasoning, designers can filter the generated product substructures and incorporate the chosen ones into the *Solution View* (D), culminating in the ultimate structural design solution.

5.2.2 The Appearance Design Page. After completing the structure design, designers can click the *Appearance Design* button to switch to this page for appearance design (Fig. 8). Starting

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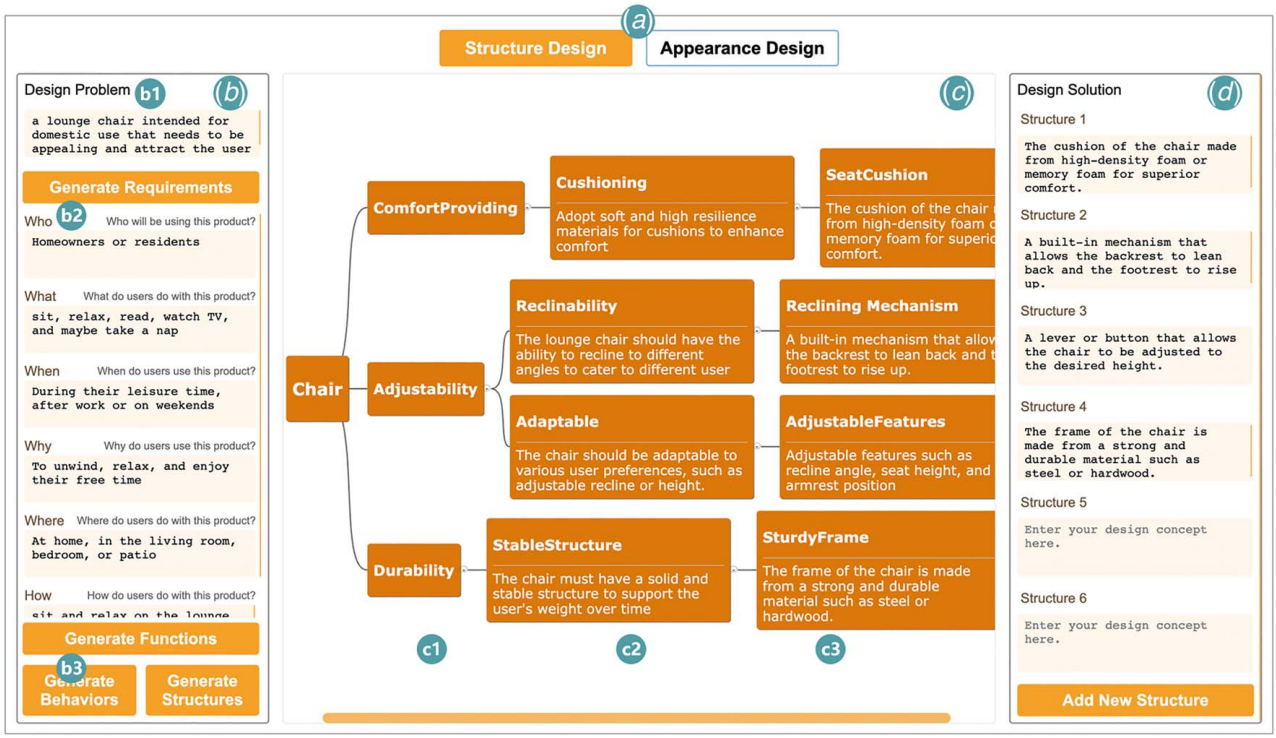


Fig. 7 An overview of the structure design interface in DesignFusion. It consists of (*a*) a Tab bar to switch between structure design and appearance design, (*b*) a *Requirement View* to input design problems and analyze functional requirements (c) a *Reasoning View* to reason about functions, behaviors, structures, and (*d*) a *Solution View* to filter and determine final structure solutions.

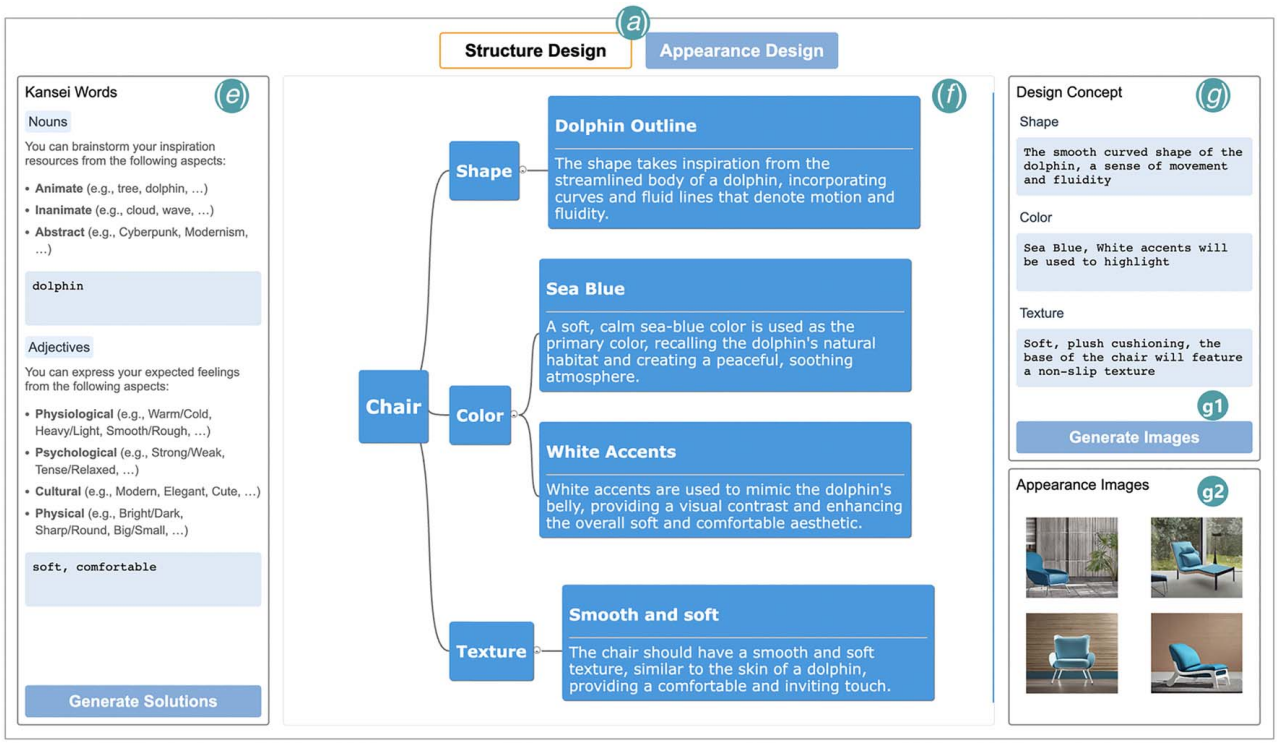


Fig. 8 An overview of the appearance design interface. It consists of (*a*) a *Tab bar* to switch between appearance design and structure design, (*e*) a *Requirement View* to input emotional requirements, (*f*) a *Reasoning View* to reason about shapes, colors, textures, and (*g*) a *Solution View* to filter and determine final appearance solutions.

from the *Requirement View* (*e*), designers conceptualize Kansei words to articulate user emotional requirements based on the answers of 5W1H and input them into the textboxes. These Kansei words can include nouns representing imagery or adjectives conveying feelings. Then, designers click the *Generate Solutions* button below, prompting the LLM to generate appearance design

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Table 3 Demographics of all the participants in user study II

ID	Gender	Age	Edu. Exp.	Use of LLMs	Use of text-to-image models
P1	Male	18–24	>5	Proficient	Proficient
P2	Female	18–24	3–5	Used before	Used before
P3	Male	18–24	3–5	Proficient	Used before
P4	Female	18–24	3–5	Proficient	Proficient
P5	Female	18–24	>5	Proficient	Proficient
P6	Male	18–24	0	Proficient	Used before
P7	Male	18–24	0	Proficient	Proficient
P8	Male	25–30	0	Used before	Never used
P9	Female	18–24	0	Used before	Never used
P10	Male	18–24	0	Proficient	Never used

Note: This table includes participants' ID, gender, age, design education experiences (Edu. Exp.), degree of familiarity with the interactive systems for LLMs (e.g., ChatGPT), and text-to-image models (e.g., stable diffusion, Midjourney).

solutions, which is structured around three key visual elements: shape, color, and texture in our approach. The LLM is responsible for generating multiple feasible values for each visual element based on the Kansei words. To represent this hierarchical solution, the *Reasoning View* (f) for appearance design also utilizes a mind map. This view also offers the same set of four operations, enabling designers to manipulate and refine each reasoning result **(O2)**. To externalizing these visual elements, designers can select one or more values for them and input them into the *Solution View* (g1). Upon clicking the *Generate Images* button, the system triggers a text-to-image model, using the selected visual elements as prompts to generate visual design images. Each click results in the generation of four images, which are displayed in the bottom of the view (g2). These images effectively combine the various visual elements within the solution and visually convey them, facilitating designers in rapid realization and assessment of the design **(O3)**. If they find the generated results unsatisfactory, they can choose to either regenerate the images or iterate on the design by retracing the reasoning process within the mind map until they reach a satisfactory visual design solution.

5.3 Implementation Details. The user interface of DesignFusion is implemented as a web-based application using HTML, CSS, and JAVASCRIPT. The creation and interaction of mind maps on it are implemented using the jsmind.js library. The back end, developed in PYTHON, handles data processing and the communication with the underlying generative models. Communication between the user interface and the back end is facilitated by an API built using the Django approach. The code is available online.[2]

Our system uses GPT-3.5-turbo [74] as the underlying LLM, with temperature = 0.7, top p = 1, and max tokens = 4, 097. For each conceptual design task, the system's back-end creates a chat to engage in multi-turn conversation with GPT-3.5. After creating the chat, the system employs context setting prompt in Sec. 3.2.1 to make GPT-3.5 act as a professional designer and work for a conceptual design task. During each reasoning step, the system obtains designer-corrected input from the user interface. It then edits prompt tailored to the current task following Eq. (1) and sends it to GPT-3.5. After receiving the generated results, the system parses and displays them on the user interface. To ensure that the text generated by GPT-3.5 aligns with the needs and is easy to parse, GPT-3.5 is directed to return structured text content in the form of the defined JSON objects within the prompts. Stable Diffusion 2.1 [75] is used as the underlying text-to-image model. During each image generation process, the system collects the description of the target product and the appearance design solution from the user interface, and incorporate them into an image generation prompt following Eq. (2). This prompt is subsequently fed into the model to generate images, which are displayed in the user interface. Specific details are available in Appendix C.

[2]https://github.com/zju-d3/DesignFusion

6 User Study II: Evaluating the User Experience During Human–Artificial Intelligence Co-Creation

To verify that our system can improve the transparency and controllability of generative models in conceptual design tasks, as well as the designer experience in human–AI co-creation, we conducted a within-subject user study, comparing DesignFusion with the state-of-the-art baseline interfaces on two design tasks.

6.1 Evaluation Method

6.1.1 Participants. In this experiment, we recruited ten participants from a university social networking site (none of them had participated in user study II), including five novice designers majoring in industrial design or mechanical engineering (three with 3–5 years of design education experience, two with more than 5 years of design education experience) and five novice designers without design education experience to assess the user experience of different designers utilizing our system. These participants also vary degrees of familiarity with AI generation tools. All the participants can communicate fluently in English. Each participant was paid $10/h, and the average time to complete the study was about 90 min.

6.1.2 Tasks and Procedure. We compared DesignFusion with a pipeline consisting of two state-of-the-art interfaces, ChatGPT and Stable Diffusion, labeled as the baseline system, in which designers used ChatGPT for conceptual design idea generation and Stable Diffusion for generating visual appearance solutions. To more accurately compare the differences between these two systems, each participant was asked to complete one design task on the baseline system and DesignFusion, respectively. At the same time, to prevent cognitive biases from repeatedly working on the same task, we selected two different but equally difficult tasks: creating a household desk lamp (Task 1) and a portable smart speaker (Task 2). The order of task execution was balanced, half of the participants used the baseline system in task 1 and DesignFusion in task 2, while the other half did the opposite. The output requirements were the same as in Study I, and each task had a maximum time limit of 30 min.

Before the experiment started, participants filled out a survey about personal information (as shown in Table 3) and received a 20-minute training on using the baseline system and prompt writing. The prompt template using by Group A in user study I was provided to participants for reference, giving them the flexibility to either utilize it or not. Subsequently, they performed the first design task using the baseline system, during which they could engage in free conversation and interaction with the system. After completing the first task, participants received an additional 10-minute training on how to use DesignFusion and were informed that the underlying generative models for this system were the same as the one used in the baseline system. They then executed the second task using DesignFusion. After completing both tasks, participants filled out a questionnaire and engage in a semi-structured

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interview. The questionnaire required participants to rate their usage experience with the two systems on various criteria, including **match goal** (the content generated by the system matches the goal), **proud** (I'm proud of the final outputs), **think through** (the system helps me think through how to accomplish the design task), **transparent** (the system is transparent on how it arrives at the final solutions, and I can track its design process/ideas), **controllable** (I feel that I could control the entire creative process with the system), **collaborative** (I feel that I'm collaborating with the system to come up with a solution, rather than the system alone producing it). These criteria were adapted from Refs. [23,27,68], aiming to reflect the effectiveness of human-AI collaboration and self-perception of humans in it. Ratings were conducted on a five-point Likert scale (1: strongly disagree, 2: disagree, 3: neutral, 4: agree, 5: strongly agree). In the semi-structured interview, we discussed the questionnaire results with participants and requested them to compare the strengths and weaknesses of both systems. Participants also expressed their needs and expectations for human–AI co-creation tools.

6.2 Result Analysis. Figure 9 displays the distribution of Likert scale responses from all participants regarding the six questions. It is evident that, compared to the baseline system, DesignFusion significantly enhances the experience of participants in human–AI collaborative creativity. To statistically analyze the ratings on each criterion, we performed a paired t-test. The results in Table 4 show that DesignFusion encouraged deeper thought on how to complete design tasks, and possessed higher levels of transparency, controllability, and collaboration. Furthermore, participants felt more proud of the concept design solutions produced in collaboration with DesignFusion. This further demonstrates that our system effectively assists novice designers in generating high-quality design solutions. The content generated by DesignFusion matched task goals better than the baseline system, but the significance on this criterion is insufficient. The rating distribution of match goal about the baseline in the figure demonstrate that participants' evaluation of this criteria is quite different, which we speculate is related to the their different familiarity with the baseline system and diverse experience in prompt writing, While the majority of participants gave highly favorable ratings to our system, recognizing its outstanding performance in various aspects.

Participants explained their ratings in the semi-structured interviews. Through analysing the results of semi-structured interviews, we can summarize the reasons for DesignFusion's superior performance in this study as follows:

6.2.1 DesignFusion is Capable of incorporating Requirements Into the Design Process. Participants provided positive feedback on the 5W1H method in DesignFusion for requirement exploration, finding that the system-generated 5W1H answers enhanced their requirement analysis depth (*P2: "The way it describes requirements gives me a visual sense and inspires me to explore more requirements"*). It also reduced the time required for initial user research (*P4: "The summarized requirements are comprehensive, which reduces the cost of initial preparations"*). Moreover, the design solutions generated in conjunction with the requirement analysis results better aligned with users' actual needs. In contrast, the baseline system using ChatGPT did not effectively consider requirements, which resulted in design solutions that did not address real problems and fail to match task goals. Although one participant with years of design education experience (*P1*) attempted to guide ChatGPT in generating requirements, the generated content was still rough and did not provide a satisfactory outcome for him.

6.2.2 The Logic Behind the Design Generation Process of DesignFusion is Clear and Aligns With the Designerly Way of Thinking. The method of DesignFusion for structure design, where it follows a logical progression from function to behavior to structure, and its structured representation in appearance design, resonate effectively with designers (*P5: "DesignFusion has stronger logic and is more in line with design habits; it can use some common design terms to describe design solutions"*). For novice designers lacking experience, this method facilitates a more comprehensive and systematic application of design thinking to problem-solving (*P7: "It provides me with a clearer design path to follow"*). In contrast, the way used by ChatGPT to directly generate solutions resemble the design capabilities of non-professionals without formal design education, failing to provide designers with insightful creative concepts (*P1: "While interacting with ChatGPT, it seems that its responses are more aligned with the skills of non-professional designers. The answers are largely predictable, offering nothing unique to inspire my ideas"*). Additionally, the use of mind maps offers a visual representation of the aforementioned logical reasoning process, enabling designers to gain a clearer understanding of its design rationale (*P1: "the use of mind maps helps me easily discern the logical connections"*).

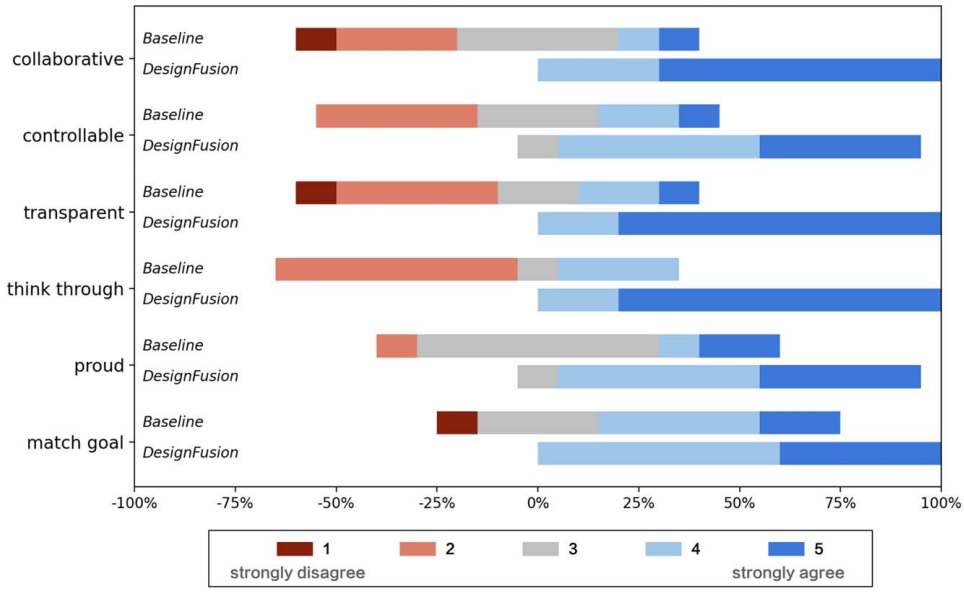


Fig. 9 Distribution of five-point Likert scale responses on the questions about human–AI co-creation across all participants for both DesignFusion and the baseline system

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Table 4 The results showing average scores, standard deviations, and *p*-values of responses on the questions about human–AI co-creation across all participants for both DesignFusion and the baseline system

Group	Collaborative	Controllable	Transparent	Think Through	Proud	Match Goal
Baseline	2.80 ± 1.14	3.00 ± 1.05	2.80 ± 1.23	2.70 ± 0.95	3.40 ± 0.97	3.60 ± 1.17
DesignFusion	$\mathbf{4.80} \pm 0.42$	$\mathbf{4.30} \pm 0.67$	$\mathbf{4.80} \pm 0.42$	$\mathbf{4.80} \pm 0.42$	$\mathbf{4.30} \pm 0.67$	$\mathbf{4.40} \pm 0.52$
p (paired t-test)	0.000^{**}	0.001^{**}	0.000^{**}	0.001^{**}	0.019^{*}	0.053

Note: **denotes $p < 0.01$ and * denotes $p < 0.05$. Bold indicates the best performance.

This also enhances the transparency of the generative models (*P6, P7, P9*).

6.2.3 DesignFusion Elevates Collaboration by Inspiring Designers to Think More Deeply. Six designers mentioned the system can guide their thought process. This is primarily because DesignFusion's step-by-step reasoning process is also a process of asking designers questions (*P1: "When interacting with ChatGPT, it's somewhat like I'm acting as a mentor, guiding it step by step to create the concept I want. But DesignFusion is the system that asks me questions and leads me to think more deeply"*). It can lead designers to think more comprehensively, focus on details they might not have considered before, and promote divergent thinking (*P1, P8, P6*), thus generating more creative ideas. This significantly increases the designers' sense of participation and collaboration throughout the human–AI co-creation (*P9*). Conversely, in ChatGPT, the generative model only attempts to answer the questions presented by designers without engaging in communication with them (*P9: "I simply input a few sentences, and ChatGPT provides an entire solution, making me feel like the solution isn't something I proposed, and it lacks the sense of participation"*). When the generated solution doesn't meet expectations, the design process often comes to a standstill (*P3: "I didn't feel that ChatGPT was helpful in my thinking process. The results it provided didn't align with my expectations, but I also didn't have specific ideas on how to modify"*).

6.2.4 The Flexible Interaction Offered by DesignFusion Enhances Controllability and Improves the Efficiency of Solution Generation. The rich and friendly interactive features provided by DesignFusion save designers' time that would otherwise be spent on frequently writing and modifying prompts, ultimately enhancing the tool's usability (*P4: "ChatGPT has a higher usage threshold than DesignFusion", P7: "ChatGPT can only interact through text prompt, which is limiting, less user-friendly, and less operable. I always have trouble making it output what I want"*). In particular, the functions to add, delete, modify, and regenerate local nodes in the mind map not only enhances user control over the output of the generative models but also accelerates the iteration of solutions, avoiding repetitive generation of the entire solution (*P3: "ChatGPT lacks step-by-step interaction, resulting in lower controllability of the entire design process"*). This provides users with a more relaxed design experience (*P3, P4*) while improving solution generation efficiency.

6.2.5 DesignFusion Expands the Boundaries of Designers' Knowledge. Most designers, especially inexperienced novices, expressed that DesignFusion provides them with a wealth of knowledge they may not possess but find highly valuable for design tasks (*P7, P8*). It also can realize the concept in designers' mind by entering Kansei words and have a rich imagination in appearance design (*P6*). Experienced designers, on the other hand, noted that compared to the standard solutions provided by ChatGPT, DesignFusion is more likely to spark fresh inspiration (*P1, P4*).

While the participants provided positive evaluations of our system's performance in the human–AI co-creation, they also provided valuable suggestions during the interviews. We will discuss these suggestions in greater detail in Sec. 7.3.

7 Discussion

As AI continues to advance in its ability to generate creative content, it is reshaping the way humans engage in creative activities. More and more teams are incorporating AI into their creative processes to provide intelligent support for human ideation. However, many questions remain to be explored in this evolving field: How can humans maximize the potential of AI in creative activities? What roles will humans play in future innovation activities? How can we control AI to generate desired outcomes? These questions open an entirely new research direction for researchers, defined as AI generated design (AIGD). This work represents a preliminary exploration and experimentation in AIGD. In this section, we discuss the implications of our results for human-AI co-creation in conceptual design, the applicability of our approach across various fields, and explore potential future directions.

7.1 Implications for Human–Artificial Intelligence Co-Creation in Conceptual Design

7.1.1 Large Language models Facilitate the Interaction Between Novice designers and the Collective Human Knowledge Base, As Well as theApplication of Knowledge. During our interviews in user study II, several novice designers mentioned that the LLM is comprehensive in conducting deep requirement and suggesting creative solutions. For instance, proposing "a lever or a rotary knob to control the angle of inclination of a chair back" illustrates this point. The generation of creativity largely depends on the knowledge of individual designers in tradition, which has its limitations. Designers often invest substantial time in knowledge retrieval. With LLMs' support, the focus of designers' work shifts from knowledge collection to learning, filtering, and integrating relevant knowledge listed by AI. Although previous methods based on semantic networks [7–9] could assist designers in rapidly retrieving knowledge, their support was notably limited. On one hand, their data sources were relatively singular, and the retrieval method based on semantic distance constrained search results to the vicinity of the search terms. On the other hand, these methods only provided explanations of knowledge. Both the initial association from design problems to relevant knowledge and the subsequent application of this knowledge required independent completion by designers. In contrast, LLMs learn and master knowledge from vast data sources [13]. They not only serve as a comprehensive and detailed knowledge repository but also complete the reasoning from problem to knowledge, and then to solution, offering support throughout the entire conceptual design process.

7.1.2 Creative Support With Artificial Intelligence Should Provide Comprehensive Assistance in Design Process and Solution Reasoning for NoviceDesigners. Previous research supporting product concept design predominantly fell into two categories: one focused on proposing abstract design process models to guide the creative activities of designers; the other sought to stimulate inspiration for specific design problems or deduce reasonable solutions. They address only one of the dual challenges novice designers face in innovative activities: how to initiate design activities or how to solve the given design problems. The traditional FBS model [31] abstracts the path to design outcomes for designers, but the specific reasoning from function to behavior, and behavior to

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structure, still requires designers to accomplish solely. The latest methods, although capable of generating textual and visual solutions through AI [15,16,76], tent to directly produce hollow results based on data distribution, lacking interpretability of the results and guidance for novice designers in the design process. Our approach integrates structured design processes and LLMs' generation capabilities, exhibiting notable efficacy in enhancing the quality of generated design solutions and effectively guiding design activities. Therefore, addressing the diverse needs of novice designers is crucial for providing creativity support for conceptual design. This involves not only offering guidance at a methodological level but also providing fine-grained reasoning assistance.

7.1.3 Cues Enhance Artificial Intelligence's Ability to Generate Innovative Ideas. In our user studies, designers encountered that sometimes the design solutions generated by AI are overly standardized and formulaic, lacking in innovation. Faced with this issue, some designers in the free conversation group attempted to stimulate the creativity of LLMs by inputting words not related to the design objective, which noticeably improved this phenomenon. For example, one designer wanted to combine the structure of the lounge chair with a plant. The LLM then suggested "designing the chair to mimic the curved shape of a banana," and the designer ultimately adopted this idea. Our proposed approach for appearance design reasoning is similar to it. The "Kansei words" conceived by the designers can be viewed as cues input into the LLM, which then provides implementation plans for shape, color, and texture. For instance, when tasked with creating "a comfortable chair with a cloud theme," the LLM offered several novel ideas, including "a cloud-shaped chair back" and "plush material." This shows that appropriate cues can enhance LLMs' ability to generate innovative ideas. Additionally, modifying the parameters of the model (i.e., "temperature" for ChatGPT) to enhance the randomness of the generated outcomes is another approach. Further empirical investigation is required to examine the influence of randomness on design ideation.

7.1.4 Synchronous Co-Creation With Artificial Intelligence Represents a Better collaboration Mode for Novice Designers Than Asynchronous Co-Creation. Although AIGD has shown tremendous potential, it also raises critical issues concerning content control and the user experience of designers [15,25]. In user study II, designers expressed that their collaborative mode with AI varied significantly between the baseline system and DesignFusion, leading to distinctly different user experiences. When using the baseline system for creation, designers felt they played the roles of leaders and decision-makers, with AI acting as an executor. This was characterized by designers posing design problems to AI, which then provided several solutions. The designers would filter and adjust these solutions or modify their input and repeat. We denote this collaboration mode as asynchronous co-creation. Based on designers' feedback, asynchronous co-creation raises the cost of solution iteration due to the generation process lacking transparency. When designers attempt to modify the input multiple times without achieving satisfactory results, they often feel frustrated and give up. Additionally, directly generating a complete solution in asynchronous co-creation can increase designers' laziness, thereby limiting their capacity for independent thought. Conversely, in creations using the DesignFusion, designers perceived that they not only led the design direction but also shared the role of executor alongside AI. They engaged in joint thinking with AI at each step of reasoning, achieving complementary results. This mode is named synchronous co-creation. Synchronous co-creation not only enhanced the designers' control and satisfaction with the final outcomes but also strengthened their sense of collaboration during the creative process. From this, we infer that in facing complex design tasks, an ideal human–AI collaboration mode is not one where humans merely make decisions. Instead, the complex task should be broken down to be easier to control, each step of reasoning should be thought and executed simultaneously by humans and AI.

7.2 Applications in Other Fields. To evaluate the scalability of our proposed approach across various fields, we conducted quick trials using DesignFusion in case studies within engineering design. Since the requirements for this field often include engineering specifications with target values, we modified our prompts to boost the applicability of the LLMs in this target domain. For instance, we substituted "product design" with "engineering design" and incorporated specific requirements, prompting the LLM to generate solutions that integrate numerical and linguistic descriptions. Moreover, as engineering design primarily emphasizes the usability of the structure, we solely assessed the structural design aspect of our approach. Details of the prompts and design problems can be found in Appendix E.

In the first design case of a rainbow sculpture [77], planned for Stockholm's harbor, the design object was a 180 deg parabolic arch spanning the harbor entrance with a 20-meter vertical clearance for small merchant ships and tall yachts. The real-design solution took comprehensive considerations of the sculpture's structure, form, stress distribution, fatigue resistance, cross-sectional shape, and manufacturing methods. A comparative analysis between the generated solution and the real solution highlighted notable consistency in terms of the sculpture's shape, material choices, and manufacturing technologies. Nonetheless, the real design delved deeper into stress analysis, considering extensive factors such as static and dynamic wind load effects, which the generated solution overlooked. Moreover, a discrepancy emerged with the generated solution erroneously altering the specified 20-meter vertical clearance to 30 m, likely a consequence of the LLM's output randomness. Despite these deviations, the generated solution exhibited innovation in analyzing cross-sectional shapes. While the real design limited its exploration to common shapes (ellipse, rectangle, triangle), ultimately selecting a triangular cross section, the generated solution proposed an airfoil shape to reduce air resistance, differing from the actual triangle but aligning well with the goal of reducing architectural drag.

In the second case of a reconfigurable obstetrics delivery bed [78], the objective was to design a bed that was easy to clean and manipulates, sturdy, low cost, and suitable for all delivery stages. The bed needed to comprise eight essential components, including back support, seat support, foot support, frame, stirrups, fluid collection system, privacy, and locking mechanisms, alongside multiple performance criteria (e.g., withstand 3000 N force, cost under $500, 10-year lifespan, collect 90% of fluids, hold up to 5 kg items). The comparison showed the shape and angle design of the simple components in the generated solution can compete with the real solution, such as rectangular back support and frames, adjustable footrests. However, it lacked consideration for complex components with special functions. For instance, the shape of seat support in the real design is rectangular with a semi-circular opening in the center connected to the fluid drainage system, while it was roughly designed as a rectangle in the generated solution, possibly due to LLM's inability to grasp inter-component relationships. For component proportions and sizes, the generated solution consistently suggested 24 in. width for related components like back and seat supports, though this was significantly smaller than the real solution's 32 in.. Some generated values showed impractical randomness, like a fluid collection system with only a 10 deg tilt. The generated solution also lacked detailed analysis of load-bearing capabilities required by the design problem, but it is consistent with the real solution in terms of the choice of supporting beams in structure. but it accurately described support beams like those in the real solution.

Overall, the generated solutions contain inherent randomness and significant errors in the specific numerical values provided, possibly due to LLM's lack of real-world physics knowledge and the ability to perform complex calculations [79]. However, DesignFusion's

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solutions offer valuable references in overall shape design, basic proportions, and manufacturing processes, indicating potential applications in engineering design. Beyond these, the classic design theories leveraged by DesignFusion, (i.e., 5W1H, the FBS model, and Kansei Engineering), has extensive real-world applications in various field, such as engineering design [80], user experience [81], and interface design [82]. Simultaneously, the LLM responsible for reasoning has acquired knowledge across diverse industries and domains. Such adaptability endows DesignFusion with significant potential to be applied into various fields.

7.3 Limitations and Future Work

7.3.1 Image Generation With Complicated Constraints. The product's appearance design incorporates multiple design elements. These are input as text constraints into a text-to-image model to guide the generation of appearance design images in our approach. However, we observed that the model struggles with reasonable integration and presentation when faced with excessive constraints. Users often need to generate multiple times to get satisfactory results. Furthermore, current text-to-image models encounter issues when visualizing complex structure design descriptions. These problems include inconsistencies between the image and the structure text description, as well as deformed structures that violate mechanical principles. In this work, we only apply the text-to-image models to generating appearance design images. Future work could focus on enhancing the image generation and text-image alignment capabilities of text-to-image models for product conceptual design tasks.

7.3.2 Comparison of Different Models and Approaches. We utilized the popular large language model GPT-3.5 and the open-source text-to-image model Stable Diffusion 2.1 as the underlying models for our experiments. But we did not test or compare other models (e.g., Llama, DALL-E 3). Moreover, we only introduced two baseline concept generation approaches for comparison with our approach in this work. Moving forward, the applicability of our approach to other generation models could be investigated and more diverse approaches of generating conceptual designs approaches could be explored.

7.3.3 Design Solution Evaluation With Artificial Intelligence. In the conceptual design process, designers typically develop multiple solutions and proceed to evaluate and filter them. The approach presented in this work primarily facilitates collaboration between designers and AI in the initial stages of requirement discovery and concept generation, with expert evaluation still being the chosen method during our user study. Although some scholars have studied to build mapping models by neural networks for the evaluation of product appearance design [83,84], these discriminative AI still require extensive manual labeling. For future work, there is potential to explore the application of pre-trained generative AI in the evaluation of design solutions, along with classic evaluation methods such as the Pugh Matrix [85], to aid designers in the evaluation, selection, and iteration of their concepts.

7.3.4 Multi-Modal Interactions. Design data exhibit in multiple modalities. However, our system only supports text input mode, making it difficult for novice designers express all types of information. In the future, voice input and hand-drawn sketches expressing their ideas can be incorporated to achieve more flexible interaction. Furthermore, AIGD can extend its output from the digital world to the physical world by integrating with 3D model generation [86–89] and 3D printing technologies [90]. This would enable rapid iteration and physical prototyping of products. Such multi-modal interaction support would significantly expand the application scope of AIGD, making it widely applicable in various creative domains.

8 Conclusions

In this study, we integrate generative models with classical design theories to introduce a novel conceptual design generation approach for product conceptual design. This approach employs the 5W1H method to enable comprehensive requirement analysis by LLMs. Subsequently, it employs the FBS model and Kansei Engineering to guide the generative models through a systematic process of structure and appearance design reasoning. The results of qualitative and quantitative results in a between-subject user study demonstrate that our approach outperforms two baseline approaches. It can conduct a more thorough requirement analysis and assist designers in generating higher-quality design solutions. To facilitate interaction between designers and our approach, we also introduce a conceptual design generation system, DesignFusion. The user interface of this system, utilizing a mind-map like layout, provides an intuitive visualization of the design generation process. The findings from a within-subject user study reveal that our system significantly enhances designers' experience in human–AI co-creation across various dimensions, such as transparency, controllability, think through, and collaboration. Finally, we discuss our initial exploration in the field of AIGD and encourage future work to explore more diversified conceptual design generation methods and multi-modal human–AI co-creation to assist designers in achieving more valuable innovations.

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Conflict of Interest

There are no conflicts of interest.

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Appendix A: Prompt Details

A.1 Prompts for Group A. Group A employed the LLM directly generate the conceptual design solutions using the following prompt, where the **[DESIGN_PROBLEM]** is provided by the designer:

Design problem: **[DESIGN_PROBLEM].** What kind of design can solve the above design problem? Please act as a professional designer and give a conceptual design of the product.

A.2 Prompts for Group C. Group C utilized our proposed conceptual design generation approach complete the conceptual design task (as illustrated in Fig. 1), which completes a complex conceptual design task by reasoning step-by-step in the designerly way of thinking using an LLM and a text-to-image model. All prompts inputted into the generative models are constructed using the three sets of prompt templates outlined in Sec. 3.2. The details are as follows:

The prompt for setting context:

Please act as a designer and work with me on a product design task. In the design process, you need to complete the structure design of the product according to FBS strategy and the appearance design based on Kansei Engineering. Please wait for my

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questions and give me the answer. Do not write all the conversation at once.

The prompt for answering the 5W1H (The **[DESIGN_PROBLEM]** is provided by the designer):

Design problem: **[DESIGN_PROBLEM]**. Please answer the following six questions according to the above design problem: 1. Who will use this product? 2. What will the user do with this product? 3. When will this product be used? 4. Where will this product be used? 5. Why do users use this product? 6. How will this product be used?

The prompt for reasoning from requirements to functions (The **[REQUIREMENTS]** is initialized with the LLM-generated answer for 5W1H in previous step and reviewed by the designer):

Product design requirement:**[REQUIREMENTS]**. What functions does the product need to have in order to meet these requirements?

The prompt for reasoning from functions to behaviors (The **[FUNCTIONS]** is initialized with the LLM-generated functions in previous step and reviewed by the designer): Product functions: **[FUNCTIONS]** What behaviors should the product implement to enable the above functions? Please use the FBS model to reason about all behaviors.

The prompt for reasoning from behaviors to structures (The **[BEHAVIORS]** is initialized with the LLM-generated behaviors in previous step and reviewed by the designer):

Behaviors that the product needs to implement: **[BEHAVIORS]** What structures can achieve the above behaviors? Please use the FBS model to reason the substructures of this product.

The prompt for reasoning from Kansei words to appearance design (The **[NOUNS]** and **[ADJECTIVES]** are the Kansei words provided by the designer based on user requirements):

Please provide a specific appearance design solution based on Kansei Engineering in terms of shape, color, and texture, ensuring that the elements of **[NOUNS]** are incorporated and **[ADJECTIVES]** emotions are conveyed.

The prompt for generating appearance images of the product (The **[SUBJECT]** is the text describes the product such as "a lounge chair," and the **[SHAPE], [COLOR], [TEXTURE]** are initialized with the LLM-generated appearance solutions in previous step and reviewed by the designer):

Product shot of **[SUBJECT]** with **[SHAPE], [COLOR], [TEXTURE]**.

Appendix B: Expert Evaluation Criteria

In user study I, we invited three professional designers as experts to evaluate the conceptual design solutions created by the participants in each group. The structure design solutions were assessed based on their novelty, functionality, and usefulness, while the appearance design solutions were evaluated for their balance, proportion, simplicity, unity, harmony, and attractiveness. Detailed descriptions of each criterion provided to the experts are shown in Tables 5 and 6.

Appendix C: Implementation Details of DesignFusion

During the process of solving a design problem, DesignFusion's backend initiates a chat and engages in multi-turn conversations with GPT-3.5 using the prompts in Appendix A. In each conversation, for the sake of facilitating the system in parsing GPT-3.5's output, we append a definition of the desired output format to the input prompt. GPT-3.5 is instructed to return structured text in the form of a JSON object. Figure 10 provides an example of the implementation code. In response to the prompt in the function code, GPT-3.5 will generate a JSON object with product behaviors as keys, each corresponding to multiple structures as values. Within the value, there is a nested JSON object. It uses structure names or keywords as keys and provides detailed structure descriptions as values. This allows us to parse the output results and display keywords and detailed descriptions of multiple structures as child nodes for the corresponding behaviors in the mind map.

Table 5 The detailed description of criterion for structure solution evaluation in user study I

Criterion	Description
Novelty	Are the functional and structural features of the solution novel?
Functionality	Do the functions implemented in the structure solution adequately fulfill user requirements?
Usefulness	Is the structure solution of the product feasible and useful?

Table 6 The detailed description of criterion for appearance solution evaluation in user study I

Criterion	Description
Balance	Are the visual design elements equally distributed to convey visual stability?
Proportion	Are the size scales between the main body and the local elements of the product harmonious?
Simplicity	Is the appearance design of the product easy to understand and free from ambiguous or complex forms?
Unity	Does the appearance design of the product reasonably gather design elements together and establish connections between elements to form a holistic design?
Harmony	Is the choice and combination of colors harmonious?
Attractiveness	Is the appearance design of the chair attractive enough for users to sit on and relax?

Appendix D: Details of the Reference Design in User Study I

Figure 11 shows the picture of the reference design, ELMER, for evaluation in user study I. The comments on ELMER from the Red Dot Award jury are as follows:

The ELMER lounge chair "achieves an almost sculptural aesthetic that is emotionally appealing and invites users to sit and relax." Intended for domestic use, the armchair owes its distinctive look to a sculptural design language inspired by balloons and other inflatable objects. The individual components of the chair follow a carefully thought-out ergonomic concept which combines a comfortably inclined backrest with generous cushioning. The biomorphic reference alludes to the Far Eastern tendency to see images in all things. Accordingly, the curved armrests are intended to resemble a loving embrace.

Appendix E: Details of the Quick Trials for Engineering Design

In section 7.2, we reported the outcomes of applying DesignFusion to two engineering design cases. To align the model's outputs with the requirements of this domain, we modified the prompt for context setting from "product design" to "engineering design." Additionally, in the prompt for reasoning from behaviors to structures, we incorporated the following requirement:

Please analyze in detail the specific parameters and indicators of the sculpture in the engineering design.

E.1 The Design Problem for Rainbow Sculpture. Conceiving the engineering design of "Rainbow," a sculpture for Stockholm's harbor. The sculpture not only had to serve as a striking visual landmark, but also adhere to strict engineering standards. You must consider the sculpture's aesthetic appeal, structural integrity, resistance to environmental factors, and manufacturability. The design must allow for vertical clearance of the vessel, integrate sprinklers and lighting, and use materials with durability and

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```python
def generate_structures(behaviors):
    prompt = f"""
        Behaviors that the product needs to achieve:
            ```{behaviors}```
        What structures can achieve the above behaviors? Please Use the
            FBS model to reason the substructures of this product.\

        I expect the output in the following JSON format:\
        {{
            "Behavior1_Name": {{
                "structure1_Name": "Corresponding Detailed structure
                    Description",
            }},
            "Behavior2_Name": {{
                "structure1_Name": "Corresponding Detailed structure
                    Description",
                "structure2_Name": "Corresponding Detailed structure
                    Description",
                ...
            }},
            ...
        }}
        I just need an output in pure JSON format. No explanations,
            comments or additional text is needed. Please follow this
            format strictly.\
        """

    messageList.append({'role': 'user', 'content': prompt})
    return messageList
```

Fig. 10 Code example of a PYTHON function that requires GPT-3.5 to generate product structures based on given product behaviors and return the output as the defined JSON object



Fig. 11 The reference design for evaluation in user study I: ELMER

strength, particularly stainless steel, to withstand wind and ice loads. Your goal is to find a balance between artistic expression and engineering feasibility, ensuring that the sculpture is both a masterpiece of design and an example of engineering. It is shown as a 180 parabolic arch spanning the harbour entrance. A vertical clearance of 20 m at middle span allows access for small commercial vessels and high-masted pleasure yachts.

E.2 The Design Problem for Reconfigurable Obstetrics Delivery Bed. The goal of this design project is to create a bed that is easy to clean and manipulates, sturdy, low cost and suitable for all delivery stages. The following is the design requirements of the project and the summary of all conditions and expectations: 1. Solid (level of importance: 5): The bed needs to bear the power of 3000 N. Suppose there are two mothers using beds at the same time, including 1.5 safety factor. 2. Safety (level of importance: 5): The product may be improperly used, so the safety factor is 1.5. 3. Affordability (level of importance: 5): Kath's labor and the annual budget of the delivery ward is about $2000, so the cost of the bed must not exceed $500. 4. Readlist (Headesium: 5): Back support requires a position settings of 180 deg (flat), 150 deg, and 130 deg to adapt to the childbirth position and provide patient comfort. The foot support also needs to be able to rotate from the flat position to help the operation of the producer during the birth. 5. Easy to clean/disinfection (importance level: 5): The bed needs to be easy to use to clean the bleach and the aqueous solution, and the cleaning time is less than 2 min. 6. Duration (level of importance: 4): The service life is 10 years. 7. Collect the body fluid (importance level: 4): The design of the bed needs to collect 90. 8. Easy to manipulate/adjust (importance level: 4): The power required for adjustable elements must be between 200 N and 300 N. 9. Comfort (level of importance: 3): The mattress must be suitable for 12 h of comfort. Privacy (level of importance: 1): Consider adding a privacy screen that can cover 50. 10. Store inspection/childbirth materials (importance level: 1): The bed needs to support the maximum 5 kg items, corresponding to the support of 75 N. The final design component includes back support, seat support, foot support, frame, stirrups, fluid collection system, privacy, and locking mechanisms.

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